

Homomodular pseudofunctors and bicategories of modules

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Introduction

The universal property for the Bénabou bicategory of distributors (although we call them "modules") presented here is somewhat implicitly spread over the series of papers [13, 14, 34, 36, 37, 4, 8] and yet, to my knowledge, does not appear in print. The inclusion of a bicategory $\mathcal{W}$ into the bicategory $\mathcal{W}\text{-Mod}$ of $\mathcal{W}$ -enriched categories and modules between them does have a completion property with respect to freely adjoining lax colimits (collages); see [37, 7].

Here we are interested in the universal property of the construction of $\mathcal{W}\text{-Mod}$ from $\mathcal{W}\text{-Cat}$. We will restrict to the case where the base $\mathcal{W}$ has but one object and indeed reduces to a monoidal category $\mathcal{V}$ so that we can refer to Kelly's book [28] for background. The usual inclusion $\mathcal{V}\text{-Cat} \rightarrow \mathcal{V}\text{-Mod}$ taking each $\mathcal{V}$ -functor to a left adjoint $\mathcal{V}$ -module has certain properties which make it a *homomodular pseudofunctor* in a sense defined herein; indeed, it is the universal such with domain $\mathcal{V}\text{-Cat}$. We also show that the inclusion $\mathcal{M}_* \rightarrow \mathcal{M}$ of the subbicategory of left adjoint morphisms into an iterative cosmos $\mathcal{M}$ (in the sense of [8]) is universal homomodular with domain $\mathcal{M}_*$.

The bicategory $\mathcal{V}\text{-Mod}$ is actually trace monoidal in the obvious bicategorical version of traced monoidal in [24]. We describe the Int construction of an autonomous monoidal bicategory from $\mathcal{V}\text{-Mod}$. The story is much the same as the sets and relations case described near the end of [24].

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1 Coslices, bipushouts and cofibrations

For functors $A \xleftarrow{f} C \xrightarrow{g} G$, we will use the notation $f \downarrow g$ for the comma category (a construction from [30] with the notation adopted in [31]); however, we prefer the term *slice of f over g* . The objects are triples $(a, fa \xrightarrow{\gamma} gb, b)$ where $a \in A$ and $b \in B$ are objects and γ is a morphism in C , the morphisms $(a, fa \xrightarrow{\gamma} gb, b) \xrightarrow{(\alpha, \beta)} (a', fa' \xrightarrow{\gamma'} gb', b')$ consist of morphisms $a \xrightarrow{\alpha} a'$ in A , $b \xrightarrow{\beta} b'$ in B such that $\gamma' \circ f\alpha = g\beta \circ \gamma$. There are projection functors $f \downarrow g \xrightarrow{s} A$ and $f \downarrow g \xrightarrow{t} B$ with an obvious natural transformation $f \circ s \xrightarrow{\lambda} g \circ t$. Restricting the objects $(a, fa \xrightarrow{\gamma} gb, b)$

of $f \downarrow g$ to those with γ invertible, we obtain the pseudopullback (known also as the “iso-comma category”).

Slices can be defined in bicategories and the results are much as for the 2-category case considered in [34]. In any bicategory $\mathcal{K}$, the *slice (or comma object)* of a cospan $A \xrightarrow{f} C \xleftarrow{g} B$ is a diagram

$$\begin{array}{ccc} f \downarrow g & \xrightarrow{t} & B \\ s \downarrow & \xRightarrow{\lambda_{f,g}} & \downarrow g \\ A & \xrightarrow{f} & C \end{array} \quad (1.1)$$

with the property that pasting the diagram defines an equivalence of categories

$$\mathcal{K}(X, f \downarrow g) \simeq \mathcal{K}(X, f) \downarrow \mathcal{K}(X, g) .$$

If in this definition we had required $\lambda_{f,g}$ to be invertible and replaced the comma category $\mathcal{K}(X, f) \downarrow \mathcal{K}(X, g)$ by the pseudopullback, we would have the definition of *bipullback* of the cospan.

Here is Proposition 5 of [34].

Proposition 1.1. *In the slice (1.1), if $f \dashv u$ then t has a right adjoint v with invertible counit. Moreover, the mate $\bar{\lambda} : s \circ v \rightarrow u \circ g$ of λ is invertible.*

$$\begin{array}{ccc} f \downarrow g & \xleftarrow{v} & B \\ s \downarrow & \xRightarrow[\cong]{\bar{\lambda}} & \downarrow g \\ A & \xleftarrow{u} & C \end{array} \quad (1.2)$$

A *coslice* of a span $A \xleftarrow{u} C \xrightarrow{v} B$ is a diagram

$$\begin{array}{ccc} C & \xrightarrow{u} & A \\ v \downarrow & \xleftarrow{\gamma} & \downarrow i \\ B & \xrightarrow{j} & u \uparrow v \end{array} \quad (1.3)$$

which is a slice in $\mathcal{K}^{\text{op}}$. Similarly, *bipushout* is bipullback in $\mathcal{K}^{\text{op}}$.

$$\begin{array}{ccccc} C & \xrightarrow{u} & A & \xrightarrow{n} & E \\ v \downarrow & \xleftarrow{\gamma} & \downarrow i & \xleftarrow[\cong]{\phi} & \downarrow k \\ B & \xrightarrow{j} & D & \xrightarrow{m} & F \end{array} \quad (1.4)$$

Here is a dual of a fact appearing, for example, as Proposition 1.7 of [8].

Proposition 1.2. *In the diagram (1.4), suppose the left-hand square is a coslice. The right-hand square is a bipushout if and only if the pasted diagram is a coslice of the span $E \xleftarrow{n \circ u} C \xrightarrow{v} B$.*

The coslice $u \uparrow C$ of the span $A \xleftarrow{u} C \xrightarrow{1_C} C$ is also called the *lax colimit* or *collage* of the diagram consisting of the single morphism $C \xrightarrow{u} A$. In this case, we can also write the coslice square as a triangle (1.5).

$$\begin{array}{ccc}
 C & \xrightarrow{u} & A \\
 & \nwarrow \scriptstyle j_u & \nearrow \scriptstyle i_u \\
 & u \uparrow C &
 \end{array}
 \quad (1.5)$$

As a consequence of a dual of Proposition 1.5 below, we have:

Proposition 1.3. *Take $B = C$ and $v = 1_C$ in diagram (1.4) and suppose the left-hand square is a coslice and the right-hand square is a bipushout. If n has a left adjoint r then m has a left adjoint ℓ satisfying $\ell \circ k \cong i \circ r$. If the counit of $r \dashv n$ is invertible then so is the counit of $\ell \dashv m$.*

As bicategorical and dual versions of Corollaries 6 and 7 of [34], we have:

Proposition 1.4. *For any collage (1.5), there is a right adjoint i_u^* for i_u with invertible unit $1_B \cong i_u^* \circ i_u$ and an invertible 2-morphism $i_u^* \circ j_u \cong u$ all defined by requiring that pasting onto γ yields the canonical invertible 2-morphism $1_A \circ u \cong u$. Moreover, u has a right adjoint if and only if j_u does; in this case, the unit of the latter adjunction is invertible.*

Suppose the bicategory $\mathcal{K}$ admits slices. Each morphism $E \xrightarrow{p} A$ in $\mathcal{K}$ induces a morphism $E \downarrow E \xrightarrow{\bar{p}} A \downarrow E$ and isomorphisms $ps \cong s\bar{p}$ and $t\bar{p} \cong t$ which paste onto the 2-cell $\lambda_{A,p}$ to yield $p\lambda_{E,E}$ (where we often write an object in place of its identity morphism). The morphism $E \xrightarrow{p} A$ will be called a *fibration* (although it should properly be called a “pseudofibration”) when $\bar{p}$ has a right adjoint with invertible counit in the bicategory $\mathcal{K}$. (In case $\mathcal{K}$ is a 2-category it makes sense to ask that the counit should be an identity; we then have a *strict fibration*.)

$$\begin{array}{ccccc}
 E \downarrow E & \xrightarrow{\bar{p}} & A \downarrow E & \xrightarrow{t} & E \\
 \downarrow s & \xRightarrow{\cong} & \downarrow s & \xRightarrow{\lambda_{A,p}} & \downarrow p \\
 E & \xrightarrow{p} & A & \xrightarrow{1_A} & A
 \end{array}
 \quad (1.6)$$

If the bicategory $\mathcal{K}$ does not admit the appropriate slices then a morphism in $\mathcal{K}$ is a *fibration* when it is taken to a fibration by the bicategorical Yoneda embedding $\mathfrak{y} : \mathcal{K} \rightarrow \text{Hom}(\mathcal{K}^{\text{op}}, \text{Cat})$.

Fibrations have the property that pullbacks along them are automatically pseudopullbacks [22] and hence bipullbacks in $\mathbf{Cat}$.

An *opfibration* is a fibration in $\mathcal{K}^{\text{co}}$. A *cofibration* is a fibration in $\mathcal{K}^{\text{op}}$. A *coopfibration* is a fibration in $\mathcal{K}^{\text{coop}}$.

If $\mathcal{K}$ admits coslices, a morphism $C \xrightarrow{u} A$ is a cofibration if and only if the canonical morphism $u \uparrow C \xrightarrow{\bar{u}} A \uparrow A$ induced by $\bar{u}$ has a left adjoint ℓ with invertible counit $\ell \circ \bar{u} \cong 1_{u \uparrow C}$.

$$A \uparrow A \begin{array}{c} \xrightarrow{\ell} \\ \perp \\ \xleftarrow{\bar{u}} \end{array} u \uparrow C \begin{array}{c} \xrightarrow{\ell \circ \bar{u}} \\ \Downarrow \cong \\ \xrightarrow{1_{u \uparrow C}} \end{array} u \uparrow C \quad (1.7)$$

In diagram (1.1), the morphism s is a fibration and t is an opfibration. In diagram (1.3), the morphism i is a cofibration and j is a coopfibration.

Proposition 1.5. *If the diagram (1.8) is a bipullback and p is a fibration in $\mathcal{K}$ then q is a fibration. Moreover, if f has a right adjoint u then r has a right adjoint t with the mate $q \circ t \Rightarrow u \circ p$ of θ invertible.*

$$\begin{array}{ccc} F & \xrightarrow{r} & E \\ q \downarrow & \xRightarrow[\cong]{\theta} & \downarrow p \\ C & \xrightarrow{f} & A \end{array} \quad (1.8)$$

Of course Proposition 1.5 has three duals.

2 The bicategory of modules

Here we remind the reader of some results essentially from [36, 4, 37, 8].

So that we are in the familiar setting of Kelly's book [28], let $\mathcal{V}$ denote a symmetric monoidal closed category which is compatibly locally presentable. We write $\mathcal{V}\text{-Cat}$ for the 2-category of (small) $\mathcal{V}$ -categories; the 1-morphisms are $\mathcal{V}$ -functors and the 2-morphisms are $\mathcal{V}$ -natural transformations.

We write $\mathcal{V}\text{-Mod}$ for the bicategory of (small) $\mathcal{V}$ -categories; the hom-category $\mathcal{V}\text{-Mod}(B, A)$ is the underlying category of the $\mathcal{V}$ -functor $\mathcal{V}$ -category $[A^{\text{op}} \otimes B, \mathcal{V}]$ whose objects we call $(\mathcal{V}\text{-})$ modules from B to A . Composition of modules $C \xrightarrow{n} B \xrightarrow{m} A$ is defined by the coend

$$(m \circ n)(a, c) = \int^b n(b, c) \otimes m(a, b) . \quad (2.9)$$

There is an inclusion pseudofunctor

$$(-)_* : \mathcal{V}\text{-Cat} \rightarrow \mathcal{V}\text{-Mod} \quad (2.10)$$

which is the identity on objects and locally fully faithful on homs; it takes the $\mathcal{V}$ -functor $B \xrightarrow{f} A$ to the module $B \xrightarrow{f_*} A$ defined by

$$f_*(a, b) = A(a, fb) . \quad (2.11)$$

There is an adjunction $f_* \dashv f^*$ in the bicategory $\mathcal{V}\text{-Mod}$ where

$$f^*(b, a) = A(fb, a) . \quad (2.12)$$

The inclusion $(-)_* : \mathcal{V}\text{-Cat} \rightarrow \mathcal{V}\text{-Mod}$, apart from a size problem, has a right pseudo-adjoint $B \mapsto \mathcal{P}B = [B^{\text{op}}, \mathcal{V}]$ which suffices, in the usual way, to prove:

Proposition 2.1. *The pseudofunctor (2.11) preserves bicategorical colimits.*

In particular, $(-)_*$ preserves coslices so a dual of Proposition 1.1 yields:

Corollary 2.2. *If the diagram (1.3) is a coslice in $\mathcal{V}\text{-Cat}$ then the mate $\bar{\gamma}$ of the image of the 2-morphism γ under the pseudofunctor (2.11) is invertible.*

$$\begin{array}{ccc} B & \xrightarrow{j_*} & u \uparrow v \\ v^* \downarrow & \xRightarrow[\cong]{\bar{\gamma}} & \downarrow i^* \\ C & \xrightarrow{u_*} & A \end{array}$$

For a module $B \xrightarrow{m} A$, the coslice $m \uparrow B$ in $\mathcal{V}\text{-Mod}$ is the $\mathcal{V}$ -category constructed in [36] as follows. The set of objects is the disjoint union of the set of objects of A and the set of objects of B . The hom $\mathcal{V}$ -objects of $m \uparrow B$ are defined by

$$(m \uparrow B)(x, y) = \begin{cases} A(x, y) & \text{if } x, y \in A \\ B(x, y) & \text{if } x, y \in B \\ m(x, y) & \text{if } x \in A \text{ and } y \in B \\ 0 & \text{otherwise .} \end{cases} \quad (2.13)$$

Composition is obtained from the compositions in A and B and their actions on m . We have fully faithful $\mathcal{V}$ -functors $A \xrightarrow{i_m} m \uparrow B$ and $B \xrightarrow{j_m} m \uparrow B$ given by inclusion and a diagram

$$\begin{array}{ccc} B & \xrightarrow{m} & A \\ & \searrow j_{m*} & \swarrow i_{m*} \\ & m \uparrow B & \end{array} \quad \begin{array}{c} \xleftarrow{\gamma} \\ \xleftarrow{\gamma} \end{array} \quad (2.14)$$

where γ is induced by the right action of A on m . If m is of the form f_* for a $\mathcal{V}$ -functor $B \xrightarrow{f} A$ then the right adjoint i_m^* of i_{m*} is also in the image of $(-)_* : \mathcal{V}\text{-Cat} \rightarrow \mathcal{V}\text{-Mod}$ as expected since $(-)_*$ preserves coslices.

A full sub- $\mathcal{V}$ -category A of a $\mathcal{V}$ -category X is called a *sieve* (“crible” in French) when, for all $x \in X$, if $x \notin A$ then $X(x, a) = 0$ for all $a \in A$. From Gray [15]

and Section 6 of [36], we have the following considerations. The inclusions $A \hookrightarrow X$ of sieves are the strict cofibrations in $\mathcal{V}\text{-Cat}$. With such a sieve, let B be the full sub- $\mathcal{V}$ -category of X consisting of those objects $x \notin A$. Then the inclusions $A \xrightarrow{i} X \xleftarrow{j} B$ provide a collage for the module $i^* \circ j_* = X(i, j) : B \rightarrow A$ in $\mathcal{V}\text{-Mod}$. An *opsieve* is a sieve in $\mathcal{V}\text{-Cat}^{\text{co}}$; in the last sentence, j is such.

Suppose we have a span $Y \xleftarrow{f} A \xrightarrow{i} X$ in $\mathcal{V}\text{-Cat}$ with i the inclusion of a sieve. There is a commutative square

$$\begin{array}{ccc} A & \xrightarrow{i} & X \\ f \downarrow & = & \downarrow h \\ Y & \xrightarrow{k} & Q \end{array} \quad (2.15)$$

in $\mathcal{V}\text{-Cat}$ defined as follows. The objects of Q are those in the disjoint union of the objects of X not in A and the objects of Y . The hom $\mathcal{V}$ -objects of Q are defined by

$$Q(w, z) = \begin{cases} Y(w, z) & \text{if } w, z \in Y \\ \int^{a \in A} X(a, z) \otimes Y(w, fa) & \text{if } z \in X \setminus A \text{ and } w \in Y \\ 0 & \text{otherwise} . \end{cases} \quad (2.16)$$

The $\mathcal{V}$ -functor k is the inclusion while hx is x when $x \notin A$ and is fx when $x \in A$. This construction has the following well-known and easily checked properties.

Proposition 2.3. *The commutative square (2.15), where i is the inclusion of a sieve (respectively, fully faithful), is a pushout (and hence by [22] a pseudopushout) in $\mathcal{V}\text{-Cat}$. Moreover, k is also the inclusion of a sieve (respectively, fully faithful) and the canonical morphism $f_* \circ i^* \rightarrow k^* \circ h_*$ in $\mathcal{V}\text{-Mod}$ is invertible.*

Proposition 2.4. *Every cofibration $A \rightarrow X$ in $\mathcal{V}\text{-Cat}$ is equivalent under A to the inclusion of a sieve and every cofibration $C \rightarrow Y$ is equivalent under C to the inclusion of an opsieve.*

3 Inverters

Let us recall a particular kind of limit from [35]. The *inverter* of a natural transformation

$$\begin{array}{ccc} A & \xrightarrow{u} & B \\ & \theta \Downarrow & \\ A & \xrightarrow{v} & B \end{array} \quad (3.17)$$

is the full subcategory H of A consisting of the objects $a \in A$ for which the components θ_a are invertible.

$$\begin{array}{ccccc}
H & \xrightarrow{\ell} & A & \xrightarrow{u} & B \\
\downarrow h & \cong & \downarrow f & \begin{array}{c} \theta \Downarrow \\ v \end{array} \cong & \downarrow g \\
H' & \xrightarrow{\ell'} & A' & \xrightarrow{u'} & B' \\
& & & \begin{array}{c} \theta' \Downarrow \\ v' \end{array} &
\end{array} \tag{3.18}$$

From Item 2.17 of [8] we have the following.

Proposition 3.1. *In Diagram 3.18, suppose the top row is an inverter and the functor g is conservative. The left square is a pseudopullback if and only if the bottom row is an inverter.*

Now take the diagram (3.17) to be in a bicategory $\mathcal{K}$, an *inverter* of θ is a morphism $H \xrightarrow{\ell} A$ such that, for all $X \in \mathcal{K}$, the functor $\mathcal{K}(X, \ell) : \mathcal{K}(X, H) \rightarrow \mathcal{K}(X, A)$ is naturally equivalent over $\mathcal{K}(X, A)$ to the inverter of the natural transformation $\mathcal{K}(X, \theta)$.

A morphism $f \in \mathcal{K}$ is *faithful* (respectively, *conservative*) when the functor $\mathcal{K}(X, f)$ is faithful (respectively, conservative) for all $X \in \mathcal{K}$. A morphism in $\mathcal{K}$ is *fickle* (respectively, *liberal*) when it is faithful (respectively, conservative) in $\mathcal{K}^{\text{op}}$.

Write $e \perp \ell$ when the following is a pseudopullback.

$$\begin{array}{ccc}
\mathcal{K}(Y, H) & \xrightarrow{\mathcal{K}(Y, \ell)} & \mathcal{K}(Y, A) \\
\mathcal{K}(e, H) \downarrow & \xRightarrow{\cong} & \downarrow \mathcal{K}(e, A) \\
\mathcal{K}(X, H) & \xrightarrow{\mathcal{K}(X, \ell)} & \mathcal{K}(X, A)
\end{array} \tag{3.19}$$

A morphism $A \xrightarrow{\ell} B$ is *strong faithful* (respectively, *strong conservative*) when $e \perp \ell$ for all fickle (respectively, liberal) morphisms e .

Proposition 3.2. *Every inverter is strong conservative.*

Proof. Diagram 3.19 is a pseudopullback using Proposition 3.1; compare 2.17 of [8] and (4.3) of [36]. $\square$

Example 3.3. *Every sieve $C \xrightarrow{i} A$ in $\mathcal{V}\text{-Cat}$ is an inverter. To see this, let B denote the full sub- $\mathcal{V}$ -category of A consisting of the objects a not in C ; so these are the objects a with $A(a, c) = 0$ for all $c \in C$. Let N be the pushout of the inclusion $B \xrightarrow{j} A$ and $B \xrightarrow{i_B} B \uparrow B$. We have two $\mathcal{V}$ -functors $A \xrightarrow{u, v} N$ which are both the identity on C while their restrictions to B are given by the inclusions i_B and j_B respectively. There is a $\mathcal{V}$ -natural transformation $\theta : u \Rightarrow v$ with identity components on C and $\gamma : i_B \Rightarrow j_B$ on B . The inverter (and identifier) of θ is i .*

4 Homodular pseudofunctors

$$\begin{array}{ccc}
 A & \xrightarrow{i} & X \\
 j \downarrow & \Rrightarrow & \downarrow h \\
 Y & \xrightarrow{k} & Q
 \end{array} \tag{4.20}$$

Definition 4.1. A pseudofunctor $T : \mathcal{K} \rightarrow \mathcal{X}$ between bicategories is homodular when

- H1. for all inverters ℓ in $\mathcal{K}$, the morphism $T\ell$ has a right adjoint $T\ell^*$ with invertible unit;
- H2. for all bipushout squares (4.20) in $\mathcal{K}$ with i a cofibration and j a cofibration, the mate $Tj \circ Ti^* \rightarrow Tk^* \circ Th$, of the image under T of the isomorphism in the square, is invertible.

Example 4.2. Suppose X is a complete category. Then $[-, X] : \text{Cat}^{\text{op}} \rightarrow \text{CAT}$ is homodular. This is because cofibrations and coslices in Cat^{op} are of course fibrations and slices in Cat . Also, since X admits pointwise right (Kan) extensions of functors into it along fibrations between small categories, Proposition 23 of [34] applies to give the two conditions for homodularity.

Example 4.3. The pseudofunctor $(-)^* : \text{Cat}^{\text{op}} \rightarrow \text{Mod}^{\text{co}}$ is homodular (where $\text{Mod} := \text{Set-Mod}$). This follows from an exactness property in Cat : every discrete two-sided fibration (in the sense of [36]) is the slice of its coslice. Another way of obtaining Example 4.2 is then to obtain $[-, X]$ by composing this $(-)^*$ with the pseudofunctor $\text{Mod}^{\text{co}} \rightarrow \text{CAT}$ taking A to $[A, X]$ and taking the module $B \xrightarrow{m} A$ to the functor $[B, X] \xrightarrow{\tilde{m}} [A, X]$ where $\tilde{m}(f) = \int_b [m(-, b), f(b)]$.

Example 4.4. Let TOP denote the bicategory of toposes and geometric morphisms. Let LEX be the bicategory of toposes and finite-limit-preserving (= left-exact) functors. The pseudofunctor $(-)_* : \text{TOP} \rightarrow \text{LEX}^{\text{co}}$, taking each geometric morphism to its left-exact left adjoint, is homodular. This follows from Section 2 of Rosebrugh-Wood [33] which identifies cofibrations in TOP^{op} as fibrations in CAT with an extra property.

Example 4.5. Suppose $\mathcal{H}$ has bipushouts, coslices and inverters, and $S : \mathcal{H} \rightarrow \mathcal{K}$ preserves them. If $T : \mathcal{K} \rightarrow \mathcal{X}$ is homodular then so is $T \circ S : \mathcal{H} \rightarrow \mathcal{X}$.

Example 4.6. If $T : \mathcal{K} \rightarrow \mathcal{X}$ is homodular then, for any bicategory $\mathcal{A}$, so is

$$\text{Hom}(\mathcal{A}, T) : \text{Hom}(\mathcal{A}, \mathcal{K}) \rightarrow \text{Hom}(\mathcal{A}, \mathcal{X}) .$$

A pseudofunctor $F : \mathcal{X} \rightarrow \mathcal{H}$ is fully faithful when its effect $F : \mathcal{X}(X, Y) \rightarrow \mathcal{H}(FX, FY)$ on hom categories is an equivalence of categories for all $X, Y \in \mathcal{X}$.

Example 4.7. If $T : \mathcal{K} \rightarrow \mathcal{X}$ is homodular and $F : \mathcal{X} \rightarrow \mathcal{H}$ is fully faithful then the composite $F \circ T : \mathcal{K} \rightarrow \mathcal{H}$ is homodular.

Proposition 4.8. Suppose $T : \mathcal{K} \rightarrow \mathcal{X}$ is a homodular pseudofunctor where $\mathcal{K}$ admits coslices. Then, for all $B \xrightarrow{f} A$ in $\mathcal{K}$, the morphism Tf has a right adjoint Tf^* .

Proof. By a dual of Proposition 1.4, in the coslice diagram

$$\begin{array}{ccc} B & \xrightarrow{1_B} & B \\ f \downarrow & \xleftarrow{\gamma} & \downarrow i \\ A & \xrightarrow{j} & B \uparrow f \end{array},$$

the morphism i is a cofibration while j has a left adjoint j' with $j' \circ j \cong 1_A$ and $j' \circ i \cong f$. So $Tf \cong Tj' \circ Ti$ has right adjoint $Ti^* \circ Tj$. $\square$

Proposition 4.9. The universal homodular pseudofunctor with domain $\mathcal{V}\text{-Cat}$ is (2.11). That is, $(-)_* : \mathcal{V}\text{-Cat} \rightarrow \mathcal{V}\text{-Mod}$ is homodular and, for each homodular pseudofunctor $T : \mathcal{V}\text{-Cat} \rightarrow \mathcal{X}$, there is a pseudofunctor $\bar{T} : \mathcal{V}\text{-Mod} \rightarrow \mathcal{X}$ which is unique up to equivalence with the property $\bar{T} \circ (-)_* \simeq T$.

Proof. That (2.11) is homodular follows from Proposition 2.3 using that equinverters and cofibrations are fully faithful $\mathcal{V}$ -functors meaning that they have right adjoints with invertible units when moved into $\mathcal{V}\text{-Mod}$.

For each module $B \xrightarrow{m} A$, define $\bar{T}m = (Ti_m)^* \circ Tj_m$ in the notation of (2.14). This definition is forced since, in $\mathcal{V}\text{-Mod}$, we have $m \cong i_m^* \circ j_{m*}$ and $\bar{T}$ is to be a pseudofunctor.

We must see that $\bar{T}$ coherently preserves composition up to isomorphism. So consider composable modules $C \xrightarrow{n} B \xrightarrow{m} A$ and form the pushout of i_n and j_m as in the left-hand square of (4.21); see (2.15). There is a fully faithful inclusion ℓ making the right-hand diagram in (4.21) commute and having $\ell j_{mon} \cong h \circ j_n$ (this can be seen from the coslice constructions). Moreover, ℓ is an inverter: this can be shown using a construction much as for cofibrations as in Example 3.3.

$$\begin{array}{ccccc} B & \xrightarrow{j_m} & m \uparrow B & \xleftarrow{i_m} & A \\ i_n \downarrow & = & \downarrow k & = & \downarrow i_{mon} \\ n \uparrow C & \xrightarrow{h} & (j_{m*} \circ n) \uparrow C & \xleftarrow{\ell} & (m \circ n) \uparrow C \end{array} \quad (4.21)$$

We have diagram (4.22) in which the bottom composite and the right side provide the collage of $m \circ n$ where, by Proposition 1.2, the middle and right squares are

bipushouts.

$$\begin{array}{ccccccc}
C & \xrightarrow{n} & B & \xrightarrow{j_{m*}} & m \uparrow B & \xrightarrow{i_m^*} & A \\
\downarrow C & \xleftarrow{\gamma} & \downarrow i_{n*} & \xleftarrow{\cong} & \downarrow k_* & \xleftarrow{\cong} & \downarrow i_{m \circ n *} \\
C & \xrightarrow{j_{n*}} & n \uparrow C & \xrightarrow{h_*} & (j_{m*} \circ n) \uparrow C & \xrightarrow{\ell^*} & (m \circ n) \uparrow C
\end{array} \quad (4.22)$$

We now have the isomorphisms

$$\begin{aligned}
\bar{T}m \circ \bar{T}n &= (Ti_m)^* \circ Tj_m \circ (Ti_n)^* \circ Tj_n \\
&\cong (Ti_m)^* \circ (Tk)^* \circ Th \circ Tj_n \\
&\cong T(k \circ i_m)^* \circ T(h \circ j_n) \\
&\cong T(\ell \circ i_{m \circ n})^* \circ T(h \circ j_n) \\
&\cong T(i_{m \circ n})^* \circ (T\ell)^* \circ T(\ell \circ j_{m \circ n}) \\
&\cong T(i_{m \circ n})^* \circ (T\ell)^* \circ T\ell \circ Tj_{m \circ n} \\
&\cong T(i_{m \circ n})^* \circ Tj_{m \circ n} \\
&= \bar{T}(m \circ n) .
\end{aligned}$$

We must see that $\bar{T}(u_*) \cong Tu$ for any $\mathcal{V}$ -functor u which will also imply that $\bar{T}$ preserves identity morphisms up to isomorphism. We use Proposition 1.4 to obtain isomorphisms

$$\begin{aligned}
\bar{T}u_* &= (Ti_{u_*})^* \circ Tj_{u_*} \\
&\cong T(i_u^*) \circ Tj_u \\
&\cong T(i_u^* \circ j_u) \\
&= Tf .
\end{aligned}$$

Coherence for these isomorphisms can be verified. $\square$

Proposition 4.9 together with Corollary 2.2 yield:

Corollary 4.10. *For any homodular pseudofunctor $T : \mathcal{V}\text{-Cat} \rightarrow \mathcal{X}$ and coslice (1.3) in $\mathcal{V}\text{-Cat}$, the mate $Tu \circ Tv^* \implies Ti^* \circ Tj$ of $T\gamma$ is invertible.*

A pseudofunctor $E : \mathcal{K} \rightarrow \mathcal{M}$ is *essentially bijective on objects* when, for each object $X \in \mathcal{M}$, there exists an object $K \in \mathcal{K}$ which is unique up to equivalence with $EK \simeq X$.

Proposition 4.11. *Universal homodular pseudofunctors are essentially bijective on objects.*

Proof. Suppose $S : \mathcal{K} \rightarrow \mathcal{D}$ is a universal homodular pseudofunctor. Factor it as $S \simeq F \circ E$ where $F : \mathcal{M} \rightarrow \mathcal{D}$ is fully faithful and $E : \mathcal{K} \rightarrow \mathcal{M}$ is the identity

on objects. By Example 4.7, E is homomodal. Since S is universal, there exists a pseudofunctor $R : \mathcal{D} \rightarrow \mathcal{M}$ with $R \circ S \simeq E$. Then $F \circ R \circ S \simeq 1_{\mathcal{D}} \circ S$ and the uniqueness clause for universality imply $F \circ R \simeq 1_{\mathcal{D}}$. By uniqueness of the factorization, $R \circ F \circ E \simeq E$ and $F \circ R \circ F \simeq F$ imply $R \circ F \simeq 1_{\mathcal{M}}$. So F is a biequivalence with the result that $S \simeq F \circ E$ is essentially bijective on objects. $\square$

Proposition 4.12. *Suppose $\mathcal{K}$ is a monoidal bicategory which admits pseudo-pushouts and coslices, and these are preserved by each $K \otimes -$ and $- \otimes K$. If $T : \mathcal{K} \rightarrow \mathcal{X}$ is the universal homomodal pseudofunctor with domain $\mathcal{K}$ then the monoidal structure on $\mathcal{K}$ extends to one on $\mathcal{X}$.*

Proof. (Sketch) Using Examples 4.5 and 4.6, we take the tensor product of $\mathcal{K}$ in the form of a pseudofunctor $\hat{\otimes} : \mathcal{K} \rightarrow \text{Hom}(\mathcal{K}, \mathcal{K})$ and composes it with $\text{Hom}(\mathcal{K}, T) : \text{Hom}(\mathcal{K}, \mathcal{K}) \rightarrow \text{Hom}(\mathcal{K}, \mathcal{X})$ to give a homomodal pseudofunctor which lands in $\text{Homod}(\mathcal{K}, \mathcal{X})$. This induces a pseudofunctor $\hat{\otimes} : \mathcal{X} \rightarrow \text{Homod}(\mathcal{K}, \mathcal{X}) \simeq \text{Hom}(\mathcal{X}, \mathcal{X})$, giving the tensor product for $\mathcal{X}$. $\square$

5 The case of an iterative cosmos

Recall from Section 6 of [8] that an *iterative cosmos* is a bicategory $\mathcal{M}$ possessing

- (a) finite bicategorical coproducts;
- (b) a Kleisli object for every monad;
- (c) finite colimits in the hom categories preserved by composition on both sides;
- (d) an algebraically free monad on each endomorphism.

Write $\mathcal{M}_*$ for the subcategory of $\mathcal{M}$ with morphisms restricted to *maps*; that is, those morphisms f with right adjoints f^* . By Proposition 6.17 of [8], if $\mathcal{M}$ is an iterative cosmos then $\mathcal{M}_*$ has bipushouts and these are preserved by the inclusion $\mathcal{M}_* \xrightarrow{\mathcal{I}} \mathcal{M}$. It then follows that $\mathcal{M}_*$ is finitely bicategorically cocomplete since $\mathcal{M}$ also has finite collages (see Subsection 6.5 and the first paragraph of the proof of Theorem 6.18 of [8] based on Proposition 2.2(a) of [37]). In fact $\mathcal{M}_*$ is a modulated bicategory in the sense of [8] so that f is strong conservative iff it is an inverter iff the unit of $f \dashv f^*$ in $\mathcal{M}$ is invertible.

Lemma 5.1. *Suppose the bicategory $\mathcal{M}$ is an iterative cosmos. If $C \xrightarrow{u} A$ is a cofibration in $\mathcal{M}_*$ then the unit of $u \dashv u^*$ is invertible.*

Proof. We will use the fact that, for a morphism v with left adjoint ℓ and right adjoint v^* in a bicategory $\mathcal{K}$, the counit of $\ell \dashv v$ is invertible if and only if the unit of $v \dashv v^*$ is invertible. This follows from a representable argument using the fact that, for $\mathcal{K} = \text{Cat}$, both conditions are equivalent to full faithfulness of v .

Since $C \xrightarrow{u} A$ is a cofibration, we are in the situation of (1.7): that is, $u \uparrow C \xrightarrow{\bar{u}} A \uparrow A$ has a left adjoint with invertible counit. By the previous paragraph, the unit for $\bar{u} \dashv \bar{u}^*$ is invertible. By Proposition 1.4, the units for $j_u \dashv j_u^*$ and $j_A \dashv j_A^*$ are both invertible. The result now follows from the isomorphism $j_A \circ u \cong \bar{u} \circ j_u$. $\square$

Proposition 5.2. *Suppose the bicategory $\mathcal{M}$ is an iterative cosmos. Then the inclusion $\mathcal{J} : \mathcal{M}_* \rightarrow \mathcal{M}$ is the universal homomodal pseudofunctor out of $\mathcal{M}_*$.*

Proof. Consider diagram (4.20) in $\mathcal{M}_*$. Using Lemma 5.1, we can apply Proposition 6.8 of [8] for the construction of the bipushout (4.20) and to obtain that it is preserved by $\mathcal{M}_* \xrightarrow{\mathcal{J}} \mathcal{M}$. Moreover, in the proof of that Proposition 6.8 in the displayed isomorphism before Theorem 6.9, we see that $j \circ i^* \Rightarrow k^* \circ h$ is invertible. So $\mathcal{M}_* \xrightarrow{\mathcal{J}} \mathcal{M}$ is homomodal. The remainder proceeds essentially as in the proof of Proposition 4.9 using Proposition 1.3 to obtain the properties of ℓ . $\square$

Proposition 5.3. *(Maybe?) Suppose the bicategory $\mathcal{M}$ is an iterative cosmos and*

$$\begin{array}{ccc} & \xrightarrow{\mathcal{Q}} & \\ \mathcal{K} & \xrightleftharpoons[\mathcal{R}]{\perp} & \mathcal{M}_* \end{array} \quad (5.23)$$

is a biadjunction with counit an equivalence $\mathcal{Q} \circ \mathcal{R} \simeq 1_{\mathcal{M}_}$. If $\mathcal{Q}$ takes equinverters to strong conservatives then*

$$\mathcal{J} = \mathcal{J}\mathcal{Q} : \mathcal{K} \rightarrow \mathcal{M}$$

is the universal homomodal pseudofunctor out of $\mathcal{K}$.

Proof. As a left biadjoint, $\mathcal{Q}$ preserves cofibrations, so $\mathcal{J}$ takes cofibrations to left adjoints because $\mathcal{J}$ does by Proposition 5.2, so the first item of Definition 4.1 holds. By assumption $\mathcal{Q}$ takes equinverters to strong conservatives which are inverters in $\mathcal{M}_*$, so item 2 holds by Proposition 5.2. Since the bipushout (4.20) is preserved by $\mathcal{Q}$, we have item 3 by Proposition 5.2. $\square$

6 The Int construction for modules

In Section 6 of [24], we constructed a symmetric autonomous monoidal category IntRel from the category Rel of sets and relations with disjoint union of sets for the tensor. A similar construction can be made starting with the bicategory $\mathcal{V}\text{-Mod}$; we will denote the resultant autonomous monoidal bicategory (in the sense of [11]) by $\mathcal{V}\text{iMod}$.

We begin by recalling that coproduct $A + B$ of $\mathcal{V}$ -categories is direct sum in $\mathcal{V}\text{-Mod}$; it is both product and coproduct:

$$\begin{aligned} \mathcal{V}\text{-Mod}(A + B, C) &\simeq \mathcal{V}\text{-Mod}(A, C) \times \mathcal{V}\text{-Mod}(B, C) \\ \mathcal{V}\text{-Mod}(C, A + B) &\simeq \mathcal{V}\text{-Mod}(C, A) \times \mathcal{V}\text{-Mod}(C, B) . \end{aligned}$$

The inclusion modules are $A \xrightarrow{\text{in}_1*} A + B \xleftarrow{\text{in}_2*} B$ and the projection modules are $A \xleftarrow{\text{in}_1^*} A + B \xrightarrow{\text{in}_2^*} B$ where $A \xrightarrow{\text{in}_1} A + B \xleftarrow{\text{in}_2} B$ are the inclusion $\mathcal{V}$ -functors. Given a module $R : B_1 + B_2 \rightarrow A_1 + A_2$, we obtain a matrix

$$\begin{bmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{bmatrix}$$

of modules where $R_{ij} = (B_j \xrightarrow{\text{in}_i*} B_1 + B_2 \xrightarrow{R} A_1 + A_2 \xrightarrow{\text{in}_i^*} A_i)$.

The free monad $R^\odot$ on an endomodule $R : U \rightarrow U$ is defined by the geometric series

$$R^\odot = \sum_{n=0}^{\infty} R^{\circ n} = U + R + R \circ R + R \circ R \circ R + \dots$$

which works because module composition preserves local coproducts. Each module morphism $\rho : R \rightarrow T$ functorially induces a monad morphism $\rho^\odot : R^\odot \rightarrow T^\odot$.

Here is the appropriate version of Lemma 6.1 of [24].

Lemma 6.1. *For modules $R, P : U \rightarrow V$ and $S : V \rightarrow U$, there are canonical isomorphisms:*

$$(i) \quad (R + S)^\odot \cong (R^\odot \circ S)^\odot \circ R^\odot \text{ for } U = V;$$

$$(ii) \quad (R \circ S)^\odot \cong U + R \circ (S \circ R)^\odot \circ S;$$

$$(iii) \quad R^\odot \cong U + R \circ R^\odot \text{ for } U = V;$$

$$(iv) \quad R^\odot \circ R \cong R \circ R^\odot \text{ for } U = V;$$

$$(v) \quad (R \circ S)^\odot \circ R \cong R \circ (S \circ R)^\odot;$$

$$(vi) \quad (U \xrightarrow{0} U)^\odot \cong (U \xrightarrow{U} U).$$

This brings us to the definition of the bicategory $\mathcal{V}\text{iMod}$. The objects are pairs (X, U) of $\mathcal{V}$ -categories. Morphisms $R : (X, U) \rightarrow (Y, V)$ are modules $X + V \rightarrow Y + U$ with matrices

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

depicted by diagrams of modules as in (6.24).

$$\begin{array}{ccc} X & \xrightarrow{A} & Y \\ c \downarrow & & \uparrow B \\ U & \xleftarrow{D} & V \end{array} \quad (6.24)$$

Composition of morphisms is as shown in diagram (6.25) (where the composition symbol $\circ$ is omitted to save space).

$$\begin{array}{ccc}
\begin{array}{ccccc}
X & \xrightarrow{A} & Y & \xrightarrow{E} & Z \\
\downarrow C & & \downarrow B & & \downarrow F \\
U & \xleftarrow{D} & V & \xleftarrow{H} & W
\end{array} & = & \begin{array}{ccc}
X & \xrightarrow{E(BG)^\circ A} & Z \\
\downarrow C+D(GB)^\circ GA & & \downarrow F+E(BG)^\circ BH \\
U & \xleftarrow{D(GB)^\circ H} & W
\end{array}
\end{array} \quad (6.25)$$

The identity morphism of (X, U) is $\begin{bmatrix} X & 0 \\ 0 & U \end{bmatrix}$.

The 2-morphisms $\alpha : R \Rightarrow S : (X, U) \rightarrow (Y, V)$ are modules $\alpha : R \Rightarrow S : X + V \rightarrow Y + U$; and these can be written as 2×2 matrices of 2-cells between the entries of the matrices for R and S . The associativity equivalences are obtained using Lemma 6.1.

The autonomous monoidal structure on $\mathcal{V}\text{iMod}$ has the symmetric tensor product given on objects by

$$(X, U) \boxplus (X', U') = (X + U, X' + U')$$

and the dual given by $(X, U)^* = (U, X)$ with counit as in diagram (6.26).

$$\begin{array}{ccc}
X + U & \xrightarrow{\quad} & 0 \\
\begin{bmatrix} 0 & U \\ X & 0 \end{bmatrix} \downarrow & & \uparrow \\
U + X & \xleftarrow{\quad} & 0
\end{array} \quad (6.26)$$

There is another monoidal structure on $\mathcal{V}\text{iMod}$ defined on objects by

$$(X, U) \boxtimes (X', U') = (X \otimes X' + U \otimes U', X \otimes U' + U \otimes X') .$$

The inclusion pseudofunctor $\mathcal{V}\text{-Mod} \rightarrow \mathcal{V}\text{iMod}$ taking X to $(X, 0)$ is strong monoidal in that it takes $\otimes$ to $\boxtimes$.

7 Stuff types and clubs

This section reviews the ideas of Kelly [25, 26], Joyal [19, 21], Baez-Dolan [2], and Byrne [6].

We denote the groupoid of finite sets and bijective functions by $\mathfrak{E}$. In the terminology of Baez-Dolan [2], a *stuff type* is a functor $F : \mathcal{X} \rightarrow \mathfrak{E}$ where $\mathcal{X}$ is a groupoid. We will write $\mathcal{X}_n$ for the full subgroupoid of $\mathcal{X}$ consisting the objects X such that FX has cardinality $|FX| = n \in \mathbb{N}$. We have $\mathcal{X} = \sum_{n=0}^{\infty} \mathcal{X}_n$.

Write $\mathfrak{S}$ for the symmetric groupoid: the objects are the natural numbers $n \in \mathbb{N}$ and the homset $\mathfrak{S}(m, n)$ is empty unless $m = n$ in which case it consists of

permutations of $\{1, 2, \dots, n\}$; composition is permutation multiplication. Regard the symmetric group $\mathfrak{S}_n$ as the full subcategory of $\mathfrak{S}$ with one object n .

A category is *skeletal* when every isomorphism is an automorphism. Every category is equivalent to a skeletal category. The groupoid $\mathfrak{S}$ is skeletal and $\mathfrak{E}$ is equivalent to it.

For a skeletal groupoid $\mathcal{X}$, define its *cardinality* by

$$|\mathcal{X}| = \sum_{X \in \mathcal{X}} \frac{1}{|\mathcal{X}(X, X)|} \quad (7.27)$$

and the cardinality of any groupoid is the cardinal of any equivalent skeleton; see [2, 6]. This is just a formal series of rational numbers unless it actually converges in $\mathbb{R}$. An important case is:

$$|\mathfrak{E}| = |\mathfrak{S}| = \sum_{n \in \mathfrak{S}} \frac{1}{|\mathfrak{S}_n|} = e . \quad (7.28)$$

The 2-category of groupoids is denoted by $\mathbf{Gpd}$. For any functor $A : \mathcal{X} \rightarrow \mathbf{Gpd}$, let $P_A : \mathcal{E}A \rightarrow \mathcal{X}$ be the corresponding opfibration: morphisms

$$(X, a \in AX) \xrightarrow{(f, \alpha)} (Y, b \in AY)$$

of $\mathcal{E}A$ consist of $X \xrightarrow{f} Y$ in $\mathcal{X}$ and $(Af)a \xrightarrow{\alpha} b$ in AY ; and $P_A(X, a) = X$, $P_A(f, \alpha) = f$. For skeletal $\mathcal{X}$, we have

$$|\mathcal{E}A| = \sum_{X \in \mathcal{X}} \frac{|AX|}{|\mathcal{X}(X, X)|} . \quad (7.29)$$

In particular, if we take A to be the composite of a stuff type $\mathcal{X} \rightarrow \mathfrak{E}$ with the inclusion $\mathfrak{E} \hookrightarrow \mathbf{Gpd}$, $n \mapsto \langle n \rangle = \{1, 2, \dots, n\}$, we denote the groupoid $\mathcal{E}A$ by $\mathcal{E}F$.

Taking F to be the identity of $\mathfrak{E}$, we obtain the groupoid $\mathcal{E}\mathfrak{E} := \mathcal{E}1_{\mathfrak{E}}$ of finite pointed sets $(N \ni k)$ and point-preserving bijections. So (7.29) gives

$$|\mathcal{E}\mathfrak{E}| = \sum_{n \in \mathfrak{S}} \frac{|\langle n \rangle|}{|\mathfrak{S}(n, n)|} = \sum_{n=1}^{\infty} \frac{1}{(n-1)!} = e - 1 . \quad (7.30)$$

We have a deletion functor

$$\backslash : \mathcal{E}\mathfrak{E} \rightarrow \mathfrak{E} , (N \ni k) \mapsto N \backslash k . \quad (7.31)$$

The *derivative* of a stuff type $F : \mathcal{X} \rightarrow \mathfrak{E}$ is the stuff type

$$\partial F = \left(\mathcal{E}F \xrightarrow{\mathcal{E}(F)} \mathcal{E}\mathfrak{E} \xrightarrow{\backslash} \mathfrak{E} \right) . \quad (7.32)$$

Given a stuff type $F : \mathcal{X} \rightarrow \mathfrak{E}$ and $N \in \mathfrak{E}$, we have the slice groupoid $F \downarrow N$ (or “pseudofibre of F over N ”). This gives the object assignment for a functor $F_\downarrow : \mathfrak{E} \rightarrow \mathbf{Gpd}$. From (7.29) we have $|\mathcal{X}| = |F_\downarrow|$.

The functor F is faithful if and only if $F_\downarrow$ factors, up to equivalence, through the inclusion $\mathbf{Set} \hookrightarrow \mathbf{Gpd}$. Functors $\mathfrak{E} \rightarrow \mathbf{Set}$ are *species of structure* in the sense of Joyal [19] and so are strict faithful stuff types.

Given a stuff type $F : \mathcal{X} \rightarrow \mathfrak{E}$ and a groupoid $\mathcal{Z}$, define the groupoid $\widehat{F}(\mathcal{Z})$ of *F-stuffed $\mathcal{Z}$ -coloured finite sets* to be the pseudoslice as shown in diagram (7.33).

$$\begin{array}{ccccc}
\widehat{F}(\mathcal{Z}) & \xrightarrow{\text{cod}} & \mathfrak{E} \downarrow \mathcal{Z} & \xrightarrow{\quad} & \mathbf{1} \\
\text{dom} \downarrow & \xRightarrow{\cong} & \text{dom} \downarrow & \xrightarrow{\quad \Lambda \quad} & \downarrow \mathcal{Z} \\
\mathcal{X} & \xrightarrow{F} & \mathfrak{E} & \xrightarrow{\text{incl}} & \mathbf{Gpd}
\end{array} \tag{7.33}$$

The objects of $\widehat{F}(\mathcal{Z})$ are $(X, FX \xrightarrow{\alpha} N \xrightarrow{Z} \mathcal{Z})$ where $X \in \mathcal{X}$, $FX \xrightarrow{\alpha} N \in \mathfrak{E}$, and $N \xrightarrow{Z} \mathcal{Z} \in \mathbf{Gpd}$. The morphisms $(\xi, \sigma, \zeta) : (X, FX \xrightarrow{\alpha} N \xrightarrow{Z} \mathcal{Z}) \rightarrow (X', FX' \xrightarrow{\alpha'} N' \xrightarrow{Z'} \mathcal{Z})$ can be depicted as in diagram (7.34).

$$\begin{array}{ccccccc}
X & & FX & \xrightarrow{\alpha} & N & \xrightarrow{Z} & \mathcal{Z} \\
\xi \downarrow & & F\xi \downarrow & & \sigma \downarrow & \searrow \zeta & \nearrow Z' \\
X' & & A_2 & \xrightarrow{\alpha'} & N' & &
\end{array} \tag{7.34}$$

We obtain a 2-functor $\widehat{F} : \mathbf{Gpd} \rightarrow \mathbf{Gpd}$ with object assignment taking $\mathcal{Z}$ to $\widehat{F}(\mathcal{Z})$. We have a pseudonatural transformation Θ as in (7.35) exhibiting $\widehat{F}$ as a left Kan pseudoextension of $F_\downarrow$ along the inclusion. The component Θ_N of Θ at N takes $(X, FX \cong N)$ to $(X, FX \cong N \xrightarrow{1_N} N)$.

$$\begin{array}{ccc}
\mathfrak{E} & \xrightarrow{\text{incl}} & \mathbf{Gpd} \\
F_\downarrow \searrow & \xrightarrow{\quad \Theta \quad} & \nearrow \widehat{F} \\
& \mathbf{Gpd} &
\end{array} \tag{7.35}$$

The 2-category of stuff types is the pseudoslice $\mathbf{Gpd} \downarrow_{\text{ps}} \mathfrak{E}$. We have a 2-functor

$$\widehat{(-)} : \mathbf{Gpd} \downarrow_{\text{ps}} \mathfrak{E} \rightarrow [\mathbf{Gpd}, \mathbf{Gpd}] \tag{7.36}$$

taking F to $\widehat{F}$. Moreover, the 2-functor $\widehat{(-)}$ is *fully faithful* (see Theorem 5.1 on page 202 of [27]). It is not an equivalence.

There is an operation of *addition* of stuff types which is taken by (7.36) to pointwise coproduct of endo-2-functors. For the binary case, the sum of $F : \mathcal{X} \rightarrow \mathfrak{E}$ and $G : \mathcal{Y} \rightarrow \mathfrak{E}$ is the induced $[F, G] : \mathcal{X} + \mathcal{Y} \rightarrow \mathfrak{E}$.

There is an operation of *multiplication* of stuff types which is taken by (7.36) to pointwise product of endo-2-functors. It uses the monoidal structure on $\mathfrak{E}$ given by coproduct (disjoint union) of sets. For the binary case, the *product* (it is not the categorical product) of $F : \mathcal{X} \rightarrow \mathfrak{E}$ and $G : \mathcal{Y} \rightarrow \mathfrak{E}$ is the composite

$$F \boxtimes G = \left(\mathcal{X} \times \mathcal{Y} \xrightarrow{F \times G} \mathfrak{E} \times \mathfrak{E} \xrightarrow{+} \mathfrak{E} \right) .$$

There is a Leibniz Rule arising from the two possibilities for $z \in FX + GY$, whether it is in FX or GY .

Proposition 7.1 (Baez-Dolan [2]). *For stuff types $F : \mathcal{X} \rightarrow \mathfrak{E}$ and $G : \mathcal{Y} \rightarrow \mathfrak{E}$, in $\mathbf{Gpd} \downarrow_{\text{ps}} \mathfrak{E}$ there is a natural equivalence*

$$\partial(F \boxtimes G) \simeq \partial F \boxtimes G + F \boxtimes \partial G .$$

There is an operation of *substitution* (or *composition*) of stuff types which is taken by (7.36) to composition of endo-2-functors. Let $F : \mathcal{X} \rightarrow \mathfrak{E}$ and $G : \mathcal{Y} \rightarrow \mathfrak{E}$ be stuff types. We define the stuff type $G \circ F : \widehat{G}(\mathcal{X}) \rightarrow \mathfrak{E}$ to be the functor taking the morphism

$$(Y, GY \xrightarrow{\beta} M \xrightarrow{X} \mathcal{X}) \xrightarrow{(\gamma, \tau, \xi)} (Y', GY' \xrightarrow{\beta'} M' \xrightarrow{X'} \mathcal{X})$$

to

$$\sum_{j \in M} FX_j \xrightarrow{\sum_{j \in M} F(\xi_j)} \sum_{j \in M} FX'_{\tau(j)} .$$

Then

$$\widehat{G \circ F} \cong \widehat{G} \circ \widehat{F} .$$

The monoids for this substitution monoidal structure are a groupoid case of Kelly's *clubs* (see [25, 26, 27] which were about a more general category case). Indeed (7.36) is strong monoidal so that clubs are taken to 2-monads on $\mathbf{Gpd}$.

For completeness we mention the Chain Rule but we may not need it.

Proposition 7.2 (Baez-Dolan [2]). *For stuff types $F : \mathcal{X} \rightarrow \mathfrak{E}$ and $G : \mathcal{Y} \rightarrow \mathfrak{E}$, in $\mathbf{Gpd} \downarrow_{\text{ps}} \mathfrak{E}$ there is a natural equivalence*

$$\partial(G \circ F) \simeq (\partial G \circ F) \boxtimes \partial F .$$

8 Application to symmetric groups

This section was inspired by Joyal's paper [20] in which he used combinatorial species to prove that homology takes the inclusion of consecutive symmetric groups to a retraction. He proved it for the universal homological functor so that it followed for any other homological functor.

Write $\mathfrak{S}$ for the symmetric groupoid: the objects are the natural numbers $n \in \mathbb{N}$ and the homset $\mathfrak{S}(m, n)$ is empty unless $m = n$ in which case it consists of permutations of $\{1, 2, \dots, n\}$; composition is permutation multiplication. Regard the symmetric group $\mathfrak{S}_n$ as the full subcategory of $\mathfrak{S}$ with one object n .

Let us recall some concepts from Joyal's papers [19, 21]. A *species (of structure)* is a functor $F : \mathfrak{S} \rightarrow \text{Set}$. The value at n is written $F[n]$ and is thought of as the set of F -structures on the set $Jn = \{1, 2, \dots, n\}$. There is a faithful inclusion functor $J : \mathfrak{S} \rightarrow \text{Set}$ taking n to $\{1, 2, \dots, n\}$.

The left Kan extension of $F : \mathfrak{S} \rightarrow \text{Set}$ along J is denoted by $F : \text{Set} \rightarrow \text{Set}$ and its value at the set X is given by the objective Taylor series

$$F(X) = \int^{n \in \mathfrak{S}} \text{Set}(Jn, X) \times F[n] \cong \sum_{n \geq 0} X^n \times_{\mathfrak{S}_n} F[n] .$$

If $T : \mathcal{A} \rightarrow \mathcal{X}$ is the terminal functor $TA = 1$ then a coend of the form $\int^A SA \times TA$ is the colimit of $S : \mathcal{A}^{\text{op}} \rightarrow \mathcal{X}$.

If $\mathcal{A}$ is a groupoid and $(-)^* : \mathcal{A} \rightarrow \mathcal{A}^{\text{op}}$ denotes the isomorphism $A \mapsto A, f \mapsto f^{-1}$ then $\int^A K(A, A)$ is the colimit of the composite

$$\mathcal{A} \xrightarrow{\Delta} \mathcal{A} \times \mathcal{A} \xrightarrow{(-)^* \times 1} \mathcal{A}^{\text{op}} \times \mathcal{A} \xrightarrow{K} \mathcal{X} .$$

That is, $\int^A K(A, A) \cong \text{colim}_A K(A^*, A)$. As a particular case,

$$F(X) \cong \sum_{n \geq 0} (X^{n^*} \times F[n]) / \mathfrak{S}_n .$$

If F is the terminal functor in this last then we have

$$\exp(X) \cong \text{colim}_{n \in \mathfrak{S}} \text{Set}(Jn, X) \cong \sum_{n \geq 0} X^{n^*} / \mathfrak{S}_n$$

for the *exponential species*.

The *derivative* F' of a species F is its composite with the endofunctor $- + 1$ of $\mathfrak{S}$ taking n to $n + 1$ and using the obvious inclusion $\mathfrak{S}_n \rightarrow \mathfrak{S}_{n+1}$ on morphisms; so $F'[n] = F[n + 1]$. This is justified by the objective form

$$F'(X) = \sum_{n \geq 0} F[n + 1] \times_{\mathfrak{S}_n} X^n$$

of the usual (abstract) formula for the derivative of a Taylor series. We have the familiar looking formulas

$$\frac{d}{dX}(F(X) + G(X)) \cong F'(X) + G'(X) ,$$

$$\frac{d}{dX}(F(X)G(X)) \cong F'(X)G(X) + F(X)G'(X) .$$

Proposition 8.1 ([20] Section 4). *Sequences of (Set)-modules $(\mathfrak{S}_{n+1} \xrightarrow{r_n} \mathfrak{S}_n)_{n \in \mathbb{N}}$ which are retractions for the inclusions $\mathfrak{S}_n \xrightarrow{i_{n*}} \mathfrak{S}_{n+1}$ in Mod are equivalent to modules $R : \mathfrak{S} \rightarrow \mathfrak{S}$ such that the triangle of modules*

$$\begin{array}{ccc} \mathfrak{S} & \xrightarrow{-+1} & \mathfrak{S} \\ \mathfrak{S}(-,-) \searrow & & \swarrow R \\ & \mathfrak{S} & \end{array}$$

commutes up to isomorphism. The equivalence is defined by

$$R[m, n] = \begin{cases} r_n[m, n] & \text{if } n = m + 1 \\ 0 & \text{otherwise} . \end{cases}$$

The triangle condition means $R[m, k + 1] \cong \mathfrak{S}(m, k)$.

The module $R : \mathfrak{S} \rightarrow \mathfrak{S}$ amounts to a functor $R : \mathfrak{S}^{\text{op}} \times \mathfrak{S} \rightarrow \text{Set}$. So we have a 2-variable species $R^\dagger : \mathfrak{S} \times \mathfrak{S} \rightarrow \text{Set}$ taking (m, n) to $R[m^*, n]$ with objective Taylor series defined by

$$R^\dagger(X, Y) = \int^{m, n} \text{Set}(Jn, X) \times \text{Set}(Jm, Y) \times R[m^*, n] .$$

Using the property of R from Proposition 20, we have

$$\begin{aligned} \frac{\partial}{\partial X} R^\dagger(X, Y) &\cong \int^{m, n} \text{Set}(Jn, X) \times \text{Set}(Jm, Y) \times R[m^*, n + 1] \\ &\cong \int^{m, n} \text{Set}(Jn, X) \times \text{Set}(Jm, Y) \times \mathfrak{S}(m^*, n) \\ &\cong \int^m \text{Set}(Jm^*, X) \times \text{Set}(Jm, Y) \\ &\cong \text{colim}_m \text{Set}(Jm, X) \times \text{Set}(Jm, Y) \\ &\cong \int^m \text{Set}(Jm, X \times Y) \\ &\cong \exp(X \times Y) . \end{aligned}$$

As motivation for the next step, suppose we have a sequence of functions $f_0, f_1, f_2, f_3, \dots$ such that f_{n+1} is the derivative of f_n for all $n \geq 0$, and $g_0, g_1, g_2, g_3, \dots$ such that g_n is the derivative of g_{n+1} for all $n \geq 0$. Then the rule for differentiating a product gives:

$$f_0(x) g_0(x) = \frac{d}{dx} \sum_{k \geq 0} (-1)^k f_k(x) g_{k+1}(x) .$$

In particular, we obtain a Taylor series for an antiderivative of $f(x)$ in terms of its derivatives by taking $f_0(x) = f(x)$ and $g_n(x) = \frac{x^n}{n!}$:

$$\begin{aligned} \int f(x)dx &= \sum_{k \geq 0} (-1)^k f^{(k)}(x) \frac{x^{k+1}}{(k+1)!} \\ &= f(x)x - f'(x)\frac{x^2}{2!} + f''(x)\frac{x^3}{3!} - \dots \end{aligned}$$

Deprived of negatives, we can write this as

$$\int f(x)dx + \sum_{k \text{ odd}} f^{(k)}(x) \frac{x^{k+1}}{(k+1)!} = \sum_{k \text{ even}} f^{(k)}(x) \frac{x^{k+1}}{(k+1)!} .$$

Similarly, an antiderivative F^J for a species F would satisfy the formula

$$F^J(X) + \sum_{k \text{ odd}} (X^{k+1}/\mathfrak{S}_{k+1}) \times F^{(k)}(X) \cong \sum_{k \text{ even}} (X^{k+1}/\mathfrak{S}_{k+1}) \times F^{(k)}(X) .$$

9 Commutative monoids

The reason Kelly's paper [29] on May's operads [32] was not published properly in the early 1970s was that Mac Lane, as editor, asked the author to consider adding material explaining the connection between operads and Kelly's clubs [25, 26]. I hope, in part, to clarify that connection in the next couple of sections.

The present section is essentially an idiosyncratic review of Kelly [29].

Let $\mathcal{V}$ denote a symmetric closed monoidal cocomplete category; the unit object is denoted by I and the tensor product by $\otimes$. For any set S and object V of $\mathcal{V}$, we write $S \cdot V$ for the coproduct of S copies of V :

$$\mathcal{V}(S \cdot V, W) \cong \mathcal{V}(V, W)^S .$$

We write $\mathfrak{S}$ for the symmetric groupoid; it is a skeleton of the category of finite sets and bijective functions.

Since we are assuming $\mathcal{V}$ symmetric, there is an action of $\mathfrak{S}$ on $\mathcal{V}$ which we can regard as a functor

$$(-)^{\otimes -} : \mathcal{V} \longrightarrow [\mathfrak{S}^{\text{op}}, \mathcal{V}]$$

taking X to $n \mapsto X^{\otimes n}$.

For functors $F : \mathcal{C} \rightarrow \mathcal{V}$ and $G : \mathcal{D} \rightarrow \mathcal{V}$, we write $F \underline{\otimes} G : \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{V}$ for the functor defined by

$$(F \underline{\otimes} G)(C, D) = FC \otimes GD .$$

There is also the *pointwise tensor product* $M \otimes N$ of functors $M, N : \mathcal{A} \rightarrow \mathcal{V}$ defined by

$$(M \otimes N)A = MA \otimes NA .$$

The objects of $[\mathfrak{S}, \mathcal{V}]$ are called *species of $\mathcal{V}$ -structure* after Joyal [19]. There is a Day convolution monoidal structure [9] on $[\mathfrak{S}, \mathcal{V}]$, which is called the *Cauchy product* by Aguiar-Mahajan [1], and we will denote it by $\bullet$:

$$T_1 \bullet T_2 = \int^{p,q} \mathfrak{S}(p+q, -) \cdot T_1 p \otimes T_2 q . \quad (9.37)$$

The integral sign with superscripts denotes the *coend* in the sense of Day-Kelly [10], but for the present article we merely require the case of ordinary categories as can be found in Mac Lane [31], for example. The universal property can be expressed as a natural isomorphism

$$[\mathfrak{S}, \mathcal{V}](T_1 \bullet T_2, U) \cong [\mathfrak{S} \otimes \mathfrak{S}, \mathcal{V}](T_1 \underline{\otimes} T_2, U+) .$$

As this convolution is also symmetric (by [9]), we obtain an action of $\mathfrak{S}$ in the form of a functor

$$(-)^{\bullet-} : [\mathfrak{S}, \mathcal{V}] \longrightarrow [\mathfrak{S}^{\text{op}}, [\mathfrak{S}, \mathcal{V}]]$$

taking T to $n \mapsto T^{\bullet n}$.

As with many convolutions throughout mathematics, this one can be transformed into a pointwise product. Indeed, the transform is provided by a functor

$$\widehat{(-)} : [\mathfrak{S}, \mathcal{V}] \longrightarrow [\mathcal{V}, \mathcal{V}] , \quad (9.38)$$

defined by a coend thus:

$$\widehat{T}X = \int^n Tn \otimes X^{\otimes n} .$$

It assigns an *analytic functor* [21] $\widehat{T}$ to each species T .

Proposition 9.1. $\widehat{T_1} \otimes \widehat{T_2} \cong \widehat{T_1 \bullet T_2}$. Indeed (9.38) is strong monoidal from the convolution structure on $[\mathfrak{S}, \mathcal{V}]$ to the pointwise tensor product on $[\mathcal{V}, \mathcal{V}]$.

Proof. By Yoneda [10], we have

$$\begin{aligned} \widehat{T_1}X \otimes \widehat{T_2}X &\cong \int^{n_1, n_2} T_1 n_1 \otimes X^{\otimes n_1} \otimes T_2 n_2 \otimes X^{\otimes n_2} \\ &\cong \int^{n_1, n_2, p} \mathfrak{S}(n_1 + n_2, p) \cdot T_1 n_1 \otimes T_2 n_2 \otimes X^{\otimes p} \cong \widehat{T_1 \bullet T_2}X , \end{aligned}$$

which is the main thing to be checked. $\square$

Moreover [29], there is another monoidal structure on $[\mathfrak{S}, \mathcal{V}]$ such that (9.38) is strong monoidal into $[\mathcal{V}, \mathcal{V}]$ under the composition strict monoidal structure. We write J for the unit and $\circ$ for the tensor product, and call this the *substitution monoidal structure* on $[\mathfrak{S}, \mathcal{V}]$; an explicit formula is

$$S \circ T = \int^r S r \otimes T^{\bullet r} . \quad (9.39)$$

This monoidal structure is left closed in the sense that $- \circ T : [\mathfrak{S}, \mathcal{V}] \rightarrow [\mathfrak{S}, \mathcal{V}]$ has a right adjoint $\{T, R\}$ defined by:

$$\{T, R\}r = \int^s [T^{\bullet s}r, Rs] .$$

Monoids in this monoidal category are the operads in the sense of Peter May [32]; we would prefer to call these *symmetric operads* (using “operad” for the easier case where $\mathfrak{S}$ is replaced by the discrete category of natural numbers). Every species T yields a symmetric operad $\{T, T\}$ called the *endomorphism operad* of T (since endo-internal-homs in closed categories are canonically monoids).

Proposition 9.2. *The functor (9.38) is strong monoidal from the substitution structure on $[\mathfrak{S}, \mathcal{V}]$ to the composition as tensor product on $[\mathcal{V}, \mathcal{V}]$.*

Proof. Using Proposition 9.1 iteratively at the fourth step, we have

$$\begin{aligned} \widehat{S \circ T}X &= \int^n (S \circ T)n \otimes X^{\otimes n} \cong \int^n \int^r Sr \otimes T^{\bullet r}n \otimes X^{\otimes n} \\ &\cong \int^r Sr \otimes \int^n T^{\bullet r} \otimes X^{\otimes n} \cong \int^r Sr \otimes \widehat{T^{\bullet r}}X = \widehat{S}(\widehat{T}X) , \end{aligned}$$

which is the main thing to be checked. $\square$

The endomorphism symmetric operad $\{T, T\}$ is taken by (9.38) to the codensity monad on $\mathcal{V}$.

We will provide some exemplary proofs of some properties of the tensor products defined above. Substitution distributes on the right over Cauchy:

Lemma 9.3. $(S \bullet T) \circ R \cong (S \circ R) \bullet (T \circ R)$

Proof. We have:

$$\begin{aligned} (S \bullet T) \circ R &= \int^r \int^{p,q} \mathfrak{S}(p+q, r) \cdot Sp \otimes Tq \otimes R^{\bullet r} \cong \int^{p,q} Sp \otimes Tq \otimes R^{\bullet(p+q)} \\ &\cong \int^p Sp \otimes R^{\bullet p} \otimes \int^q Tq \otimes R^{\bullet q} \cong (S \circ R) \bullet (T \circ R) , \end{aligned}$$

where the second step uses the coend form of the Yoneda Lemma. $\square$

Proposition 9.4. $(S \circ T) \circ R \cong S \circ (T \circ R)$

Proof. We have

$$(S \circ T) \circ R \cong \int^r Sr \otimes (T^{\bullet r} \circ R) \cong \int^r Sr \otimes (T \circ R)^{\bullet r} \cong S \circ (T \circ R) ,$$

where the first isomorphism uses the left closedness of substitution and the second isomorphism uses iteration of Lemma 9.3. $\square$

Since $\mathcal{V}$ is a cocomplete category, we have a colimit functor

$$\mathrm{colim}: [\mathfrak{S}^{\mathrm{op}}, \mathcal{V}] \rightarrow \mathcal{V} .$$

The *symmetric monoid functor* is defined as the composite functor

$$S: \mathcal{V} \xrightarrow{(-)^{\otimes -}} [\mathfrak{S}^{\mathrm{op}}, \mathcal{V}] \xrightarrow{\mathrm{colim}} \mathcal{V} . \quad (9.40)$$

Clearly

$$S = \widehat{I}! , \quad (9.41)$$

where $I! = (\mathfrak{S} \longrightarrow \mathbf{1} \xrightarrow{I} \mathcal{V})$.

Now $I!$ is clearly a symmetric operad. So S is a monad; its Eilenberg-Moore category $\mathcal{V}^S$ is isomorphic to the category $\mathrm{Monc}\mathcal{V}$ of commutative monoids in $\mathcal{V}$.

10 Lax colimits in bicategories

Following Bénabou [3] (also see [36]), we write $\mathrm{Hom}(\mathcal{M}, \mathcal{N})$ for the bicategory of pseudofunctors $\mathcal{M} \rightarrow \mathcal{N}$, pseudonatural transformations, and modifications.

We write $\mathrm{Hom}_{\mathrm{lax}}(\mathcal{M}, \mathcal{N})$ for the bicategory of pseudofunctors $\mathcal{M} \rightarrow \mathcal{N}$, lax natural transformations, and modifications.

Suppose $\mathcal{D}$ and $\mathcal{X}$ are bicategories. For pseudofunctors $J: \mathcal{D}^{\mathrm{op}} \rightarrow \mathrm{Cat}$ and $F: \mathcal{D} \rightarrow \mathcal{X}$, we write $\mathrm{coll}(J, F)$ for the *J-weighted lax colimit* of F . It is defined by a pseudonatural equivalence

$$\mathcal{X}(\mathrm{coll}(J, F), X) \simeq \mathrm{Hom}_{\mathrm{lax}}(\mathcal{D}^{\mathrm{op}}, \mathrm{Cat})(J, \mathcal{X}(F, X)) .$$

The “conical” case of this is where

$$J = (\mathcal{D}^{\mathrm{op}} \xrightarrow{!} \mathbf{1} \xrightarrow{\mathbf{1}} \mathrm{Cat})$$

is the constant pseudofunctor at the terminal category; then we simply write $\mathrm{coll}F$ for $\mathrm{coll}(J, F)$ and call it the *collage* of F [37].

On the other hand, every weighted lax colimit can be obtained as a collage

$$\mathrm{coll}(J, F) \simeq \mathrm{coll}(\mathcal{J} \xrightarrow{P} \mathcal{D} \xrightarrow{F} \mathcal{X})$$

where $P: \mathcal{J} \rightarrow \mathcal{D}$ is the fibration obtained by the Grothendieck construction from the pseudofunctor $J: \mathcal{D}^{\mathrm{op}} \rightarrow \mathrm{Cat}$.

As a particular case of a weight, we have the hom pseudofunctor

$$\mathcal{D}(-, -): \mathcal{D}^{\mathrm{op}} \times \mathcal{D} \longrightarrow \mathrm{Cat} .$$

The lax colimits for this weight are lax coends (see Bozapalides [5]). Put

$$\oint^D T(D, D) = \mathrm{coll}(\mathcal{D}(-, -), T)$$

for any pseudofunctor $T: \mathcal{D}^{\text{op}} \times \mathcal{D} \longrightarrow \mathcal{X}$. It can be seen that $\mathcal{X}(\oint^A T(A, A), X)$ is equivalent to the category of *lax cocones* from T to X . Weighted lax colimits can be obtained using lax coends and copowers:

$$\text{coll}(J, F) \simeq \oint^D JD \bullet FD .$$

The following is essentially due to Bozapalides [5].

Proposition 10.1. [Fubini Theorem] *Suppose $T: (\mathcal{D} \times \mathcal{E})^{\text{op}} \times \mathcal{D} \times \mathcal{E} \longrightarrow \mathcal{X}$ is a pseudofunctor into a bicategory $\mathcal{X}$ which admits the appropriate collages. Then $\oint^E T(D_1, E, D_2, E)$ is the value on objects $(D_1, D_2) \in \mathcal{D}^{\text{op}} \times \mathcal{D}$ of a pseudofunctor*

$$\oint^E T(-, E, -, E): \mathcal{D}^{\text{op}} \times \mathcal{D} \longrightarrow \text{Cat} ,$$

and

$$\oint^D \oint^E T(D, E, D, E) \simeq \oint^{D, E} T(D, E, D, E) .$$

11 Symmetric monoidales

Let $\mathcal{M}$ be a symmetric closed monoidal bicategory [11] which admits lax colimits. Again the tensor product is denoted by $\otimes$ and the unit object by I .

Since we are assuming $\mathcal{M}$ symmetric, there is an action of $\mathfrak{S}$ on $\mathcal{M}$ which we can regard as a pseudofunctor

$$(-)^{\otimes -}: \mathcal{M} \longrightarrow \text{Hom}(\mathfrak{S}^{\text{op}}, \mathcal{M})$$

taking X to $n \mapsto X^{\otimes n}$.

For pseudofunctors $F: \mathcal{C} \rightarrow \mathcal{M}$ and $G: \mathcal{D} \rightarrow \mathcal{M}$, we write $F \underline{\otimes} G: \mathcal{C} \otimes \mathcal{D} \rightarrow \mathcal{M}$ for the pseudofunctor defined by

$$(F \underline{\otimes} G)(C, D) = FC \otimes GD .$$

Write $P: \mathfrak{S}^{\text{op}} \times \mathfrak{S}^{\text{op}} \longrightarrow \text{Hom}(\mathfrak{S}, \mathcal{M})$ for the pseudofunctor defined by the copower

$$P(p, q) = \mathfrak{S}(p + q, -) \bullet I .$$

An object of $\text{Hom}(\mathfrak{S}, \mathcal{M})$ will be called a *genus of $\mathcal{M}$ -structure*. There is a symmetric closed monoidal structure on the bicategory $\text{Hom}(\mathfrak{S}, \mathcal{M})$, which we call the *genus Cauchy product*, and denote by $\bullet$:

$$T_1 \bullet T_2 = \text{coll}(P, T_1 \underline{\otimes} T_2) = \oint^{p, q} \mathfrak{S}(p + q, -) \bullet T_1 p \otimes T_2 q . \quad (11.42)$$

The universal property can be expressed as a pseudonatural equivalence

$$\text{Hom}(\mathfrak{S}, \mathcal{M})(T_1 \bullet T_2, U) \simeq \text{Hom}(\mathfrak{S} \times \mathfrak{S}, \mathcal{M})_{\text{lax}}(T_1 \underline{\otimes} T_2, U+) .$$

Associativity equivalences are obtained using the Fubini Theorem for lax coends.

We obtain an action of $\mathfrak{S}$ on $\text{Hom}(\mathfrak{S}, \mathcal{M})$ in the form of a pseudofunctor

$$(-)^{\bullet-} : \text{Hom}(\mathfrak{S}, \mathcal{M}) \longrightarrow \text{Hom}(\mathfrak{S}^{\text{op}}, \text{Hom}(\mathfrak{S}, \mathcal{M}))$$

taking T to $n \mapsto T^{\bullet n}$.

There is a pseudofunctor

$$\widehat{(-)} : \text{Hom}(\mathfrak{S}, \mathcal{M}) \longrightarrow \text{Hom}(\mathcal{M}, \mathcal{M}) \quad (11.43)$$

defined by a lax coend thus:

$$\widehat{T}X = \oint^n Tn \otimes X^{\otimes n} .$$

Moreover, there is a monoidal structure on $\text{Hom}(\mathfrak{S}, \mathcal{M})$ such that (11.43) is strong monoidal. We write J for the unit and $\circ$ for the tensor product, and call this the *genus substitution monoidal structure* on $\text{Hom}(\mathfrak{S}, \mathcal{M})$; an explicit formula is

$$S \circ T = \oint^r Sr \otimes T^{\bullet r} . \quad (11.44)$$

The monoidales (= pseudomonoids) for genus substitution will be called *symmetric clubs*. Why this is consistent with Kelly [26] may be revealed below.

The *symmetric monoidale pseudofunctor* is defined as the composite pseudofunctor of the $\mathfrak{S}$ -action and collage:

$$S : \mathcal{M} \xrightarrow{(-)^{\otimes-}} \text{Hom}(\mathfrak{S}^{\text{op}}, \mathcal{M}) \xrightarrow{\text{coll}} \mathcal{M} . \quad (11.45)$$

Notice that

$$S = \widehat{I}! . \quad (11.46)$$

Now $I!$ is a symmetric club. So S is a pseudomonad; its bicategory $\mathcal{M}^S$ of Eilenberg-Moore pseudoalgebras is isomorphic to the bicategory $\text{Mons} \mathcal{M}$ of symmetric monoidales in $\mathcal{M}$.

Example 11.1. Let $\mathcal{M} = \text{Mod}$ be the monoidal bicategory whose objects are small categories, morphisms are (two-sided) modules (= profunctors = distributors), and whose 2-cells are module morphisms. That is, for small categories $\mathcal{A}$ and $\mathcal{B}$, the hom categories of Mod are $\text{Mod}(\mathcal{A}, \mathcal{B}) = [\mathcal{B}^{\text{op}} \times \mathcal{A}, \text{Set}]$. We know that Mod admits small collages [37]. Since $\mathfrak{S}$ is a groupoid, pseudofunctors $T : \mathfrak{S} \longrightarrow \mathcal{M}$ are essentially functors $K : \mathcal{T} \longrightarrow \mathfrak{S}$ where the pseudofibre of K over $n \in \mathfrak{S}$ is Tn . Kelly's clubs are indeed functors $K : \mathcal{T} \longrightarrow \mathfrak{S}$ which are monoids for a substitution monoidal structure.

12 Review of Joyal species for objective combinatorics

The essential basic framework of [19] involves the following concepts.

1. $\mathfrak{S}$ is the *symmetric groupoid*: its objects are natural numbers, all morphisms are endomorphisms, while $\mathfrak{S}(n, n)$ is the *symmetric group* $\mathfrak{S}_n$ of permutations on the n symbols $1, 2, \dots, n$.
2. A *species (of structure)* is a functor $F : \mathfrak{S} \rightarrow \text{Set}$. The value at n is written $F[n]$ and is thought of as the set of F -structures on the set $Jn = \{1, 2, \dots, n\}$.
3. There is a faithful inclusion functor $J : \mathfrak{S} \rightarrow \text{Set}$ taking n to $\{1, 2, \dots, n\}$. The left Kan extension of $F : \mathfrak{S} \rightarrow \text{Set}$ along J is denoted by $F : \text{Set} \rightarrow \text{Set}$ and its value at the set X is given by the objective Taylor series

$$F(X) = \int^{n \in \mathfrak{S}} \text{Set}(Jn, X) \times F[n] \cong \sum_{n \geq 0} X^n \times_{\mathfrak{S}_n} F[n] .$$

Here are some useful facts.

- a. If $T : \mathcal{A} \rightarrow \mathcal{X}$ is the terminal functor $TA = 1$ then a coend of the form $\int^A SA \times TA$ is the colimit of $S : \mathcal{A}^{\text{op}} \rightarrow \mathcal{X}$.
- b. If $\mathcal{A}$ is a groupoid and $(-)^* : \mathcal{A} \rightarrow \mathcal{A}^{\text{op}}$ denotes the isomorphism $A \mapsto A, f \mapsto f^{-1}$ then $\int^A K(A, A)$ is the colimit of the composite

$$\mathcal{A} \xrightarrow{\Delta} \mathcal{A} \times \mathcal{A} \xrightarrow{(-)^* \times 1} \mathcal{A}^{\text{op}} \times \mathcal{A} \xrightarrow{K} \mathcal{X} .$$

That is, $\int^A K(A, A) \cong \text{colim}_A K(A^*, A)$.

- c. As a particular case of (b),

$$F(X) \cong \sum_{n \geq 0} (X^{n^*} \times F[n]) / \mathfrak{S}_n .$$

- d. If F is the terminal functor in (c) then (a) gives

$$\exp(X) \cong \text{colim}_{n \in \mathfrak{S}} \text{Set}(Jn, X) \cong \sum_{n \geq 0} X^{n^*} / \mathfrak{S}_n$$

for the *exponential species*.

Species can be differentiated.

- The derivative F' of a species F is its composite with the endofunctor of $\mathfrak{S}$ taking n to $n+1$ and using the obvious inclusion $\mathfrak{S}_n \rightarrow \mathfrak{S}_{n+1}$ on morphisms; so $F'[n] = F[n+1]$.
- This is justified by the objective form

$$F'(X) = \sum_{n \geq 0} F[n+1] \times_{\mathfrak{S}_n} X^n$$

of the usual (abstract) formula for the derivative of a Taylor series.

- We have familiar looking formulas which include

$$\begin{aligned} \frac{d}{dX}(F(X) + G(X)) &\cong F'(X) + G'(X) , \\ \frac{d}{dX}(F(X)G(X)) &\cong F'(X)G(X) + F(X)G'(X) . \end{aligned}$$

13 Virtual species

For ordinary Taylor series we have:

- i. Suppose we have a sequence of functions $f_0, f_1, f_2, f_3, \dots$ such that f_{n+1} is the derivative of f_n for all $n \geq 0$, and $g_0, g_1, g_2, g_3, \dots$ such that g_n is the derivative of g_{n+1} for all $n \geq 0$. Then the rule for differentiating a product gives:

$$f_0(x) g_0(x) = \frac{d}{dx} \sum_{k \geq 0} (-1)^k f_k(x) g_{k+1}(x) .$$

- ii. In particular, we obtain a Taylor series for an antiderivative of $f(x)$ in terms of its derivatives by taking $f_0(x) = f(x)$ and $g_n(x) = \frac{x^n}{n!}$:

$$\begin{aligned} \int f(x) dx &= \sum_{k \geq 0} (-1)^k f^{(k)}(x) \frac{x^{k+1}}{(k+1)!} \\ &= f(x)x - f'(x)\frac{x^2}{2!} + f''(x)\frac{x^3}{3!} - \dots . \end{aligned}$$

The formulas for derivative of species are enough to prove a species version of these antiderivative formulas except for one problem: minus signs. To overcome this, we move to *virtual species* [21].

- The main fact that makes virtual species work is already apparent for *virtual sets*: that is, pairs (P, M) of sets, with M finite, modulo the relation $(P, M) \sim (Q, N)$ if $P+N \cong Q+M$. Now, from a bijection $\phi : X+U \cong Y+U$

with U finite, we can construct a trace bijection $\psi : X \cong Y$ where $\psi(x) = \phi^n(x)$ where $n > 0$ gives the time $\phi^n(x) \notin U$. So we have an equivalence relation, because $(P, M) \sim (Q, N) \sim (R, L)$ implies $P + N \cong Q + M$ and $R + N \cong Q + L$, so $P + L + N \cong R + M + N$, which then gives $(P, M) \sim (R, L)$. We write $P - M$ for the equivalence class of (P, M) .

- With these ingredients, we can make sense of

$$F(X) \cong \frac{d}{dX} \sum_{k \geq 0} (-1)^k (X^{k+1} / \mathfrak{S}_{k+1}) \times F^{(k)}(X) .$$

14 Homological functors

Let $\mathbf{gpd}$ denote the category of finite groupoids and let $\mathcal{A}$ be an additive category. Here a *homological functor* $H : \mathbf{gpd} \rightarrow \mathcal{A}$ is to be equipped with a *transfer* functor $\Sigma : \mathbf{gpd}_{\text{dof}}^{\text{op}} \rightarrow \mathcal{A}$ which agrees with H on objects and is defined on the subcategory $\mathbf{gpd}_{\text{dof}}$ of $\mathbf{gpd}$ with morphisms the discrete opfibrations. The following properties are required.

- If there is a natural transformation from f to g then $Hf = Hg$.
- If the square

$$\begin{array}{ccc} P & \xrightarrow{u} & B \\ p \downarrow \lrcorner & & \downarrow q \\ A & \xrightarrow{v} & C \end{array}$$

is a pullback with q a discrete opfibration then $\Sigma_q \circ Hv = Hu \circ \Sigma_p$.

- H preserves finite coproducts: $H(A + B) \cong HA \oplus HB$, $H\emptyset \cong 0$.

Example 14.1. Let $\mathcal{C}$ be a category with finite coproducts. For $A \in \mathbf{gpd}$, form the functor category $\mathbf{H}A = [A, \mathcal{C}]$ to obtain a 2-functor $\mathbf{H} : \mathbf{gpd} \rightarrow \mathbf{Cat}_+^{\text{op}}$. For a discrete opfibration $p : X \rightarrow A$, there is a left adjoint functor $\Sigma_p \dashv \mathbf{H}p$. Composing with the Grothendieck abstraction 2-functor $K_0 : \mathbf{Cat}_+ \rightarrow \mathbf{Abgp}$ into abelian groups, we obtain a homological functor with values in the additive category $\mathbf{Abgp}^{\text{op}}$.

Example 14.2. For $p \in \mathbf{gpd}_{\text{dof}}$ and u in $\mathbf{Gpd}$ with C an abelian group, define Σ_p by $\Sigma_p(u)a = \sum_{p(x)=a} u(\alpha_* x)$ for $a \in A$ and put $HA = \pi_0[A, C]$ with pointwise addition.

$$\begin{array}{ccc} X & \xrightarrow{p} & A \\ & \searrow u & \swarrow \Sigma_p(u) \\ & C & \end{array}$$

Our strategy based on [?] is as follows:

- To construct the universal homological functor and prove the retraction result for it. Then it will hold for all homological functors since all functors preserve retracts.
- To look at an objective homological pseudofunctor from which the universal functor can be abstracted using the Grothendieck construction K_0 of an abelian group from a category with coproducts.
- We work with the 2-category cat of finite categories as far as possible rather than finite groupoids.

15 Finite coproduct completion and partial functors

Let set denote the category of finite sets. The finite coproduct completion $\text{fam}B$ of a category B with finite homs is the closure of the image of the Yoneda embedding $\mathfrak{y}_B : B \rightarrow [B^{\text{op}}, \text{set}]$ under finite coproducts. There is a bijection taking natural transformations

$$\sum_{i \in I} B(-, b_i) \rightarrow \sum_{j \in J} B(-, c_j)$$

between finite coproducts of representables to diagrams

$$\begin{array}{ccc} I & \xrightarrow{\xi} & J \\ & \searrow b \quad \xRightarrow{\beta} \quad \swarrow c & \\ & B & \end{array}$$

where $\xi : I \rightarrow J$ is a function between finite sets. We can identify $\text{fam}B$ with the category whose objects are finite families $I \xrightarrow{b} B$ of objects of B and whose morphisms are $(\xi, \beta) : b \rightarrow c$ as in the diagram. A *partial functor* between finite categories A and B is a span $A \xleftarrow{p} X \xrightarrow{f} B$ in cat where p is a discrete opfibration. There is a bicategory parf whose objects are finite categories, morphisms are partial functors, composition is by pullback of spans, and 2-cells are as in the diagram.

$$\begin{array}{ccccc} & & X & & \\ & p \swarrow & \downarrow u & \swarrow f & \\ A & \xleftarrow{p'} & X' & \xrightarrow{f'} & B \end{array}$$

Left Kan extension

$$\begin{array}{ccc} X & \xrightarrow{p} & A \\ f \downarrow & \xRightarrow{\kappa} & \downarrow R \\ B & \longrightarrow & \text{fam}B \end{array}$$

exhibits $\text{fam}B$ as the *partial functor classifier*, where $R(a) = \sum_{p(x)=a} B(-, f(x))$. The inclusion $\text{fam}B \hookrightarrow [B^{\text{op}}, \text{set}]$ shows that every partial function gives a module $R : B^{\text{op}} \times A \rightarrow \text{set}$ where $R(a)b$ becomes $R(b, a)$. This gives a fully faithful pseudofunctor $\text{parf} \rightarrow \text{mod}$ into the bicategory of finite categories and finite modules between them.

The homs $\text{parf}(A, B)$ have finite coproducts preserved by composition on both sides. Put $\text{abparf}(A, B) = K_0\text{parf}(A, B)$ to obtain an additive category abparf . Define $H : \text{gpd} \rightarrow \text{abparf}$ by $H(A \xrightarrow{f} B) = [A \xleftarrow{1_A} A \xrightarrow{f} B]$. Define $\Sigma : \text{gpd}_{\text{dof}}^{\text{op}} \rightarrow \text{abparf}$ by $H(A \xrightarrow{p} B) = [B \xleftarrow{p} A \xrightarrow{1_A} A]$.

Proposition 15.1. *$\text{abparf}(A, B)$ with this H and Σ is universal amongst homological functors.*

A bit more explicitly, restriction along H gives an equivalence between the category of additive functors $\text{abparf} \rightarrow \mathcal{A}$ and the category of homological functors $\text{gpd} \rightarrow \mathcal{A}$. So partial functors play a universal role for homological functors akin to spans for Mackey functors and polynomials for Tambara functors.

16 The retraction

- For each $n \geq 0$, suppose $\mathfrak{S}_{n+1} \xrightarrow{r_n} \mathfrak{S}_n$ is a retraction for the inclusion $\mathfrak{S}_n \xrightarrow{i_n} \mathfrak{S}_{n+1}$ in parf .
- Define a Set-module $\mathfrak{S} \xrightarrow{R} \mathfrak{S}$ by

$$R[m, n] = \begin{cases} r_n & \text{if } n = m + 1 \\ 0 & \text{otherwise} . \end{cases}$$

- The retract condition can be expressed by commutativity of the triangle

$$\begin{array}{ccc} \mathfrak{S} & \xrightarrow{-+1} & \mathfrak{S} \\ \mathfrak{S}(-, -) \searrow & & \swarrow R \\ & \mathfrak{S} & \end{array}$$

of modules. That is,

$$R[m, k + 1] = \begin{cases} \mathfrak{S}(m, k) & \text{if } m = k \\ 0 & \text{otherwise} . \end{cases}$$

- We have a 2-variable species

$$R(X, Y) = \int^{m, n} \text{Set}(Jn, X) \times \text{Set}(Jm, Y) \times R[m^*, n] .$$

- Using the property of R from the last frame, we have

$$\begin{aligned}
\frac{d}{dX} R(X, Y) &\cong \int^{m, n} \text{Set}(Jn, X) \times \text{Set}(Jm, Y) \times R[m^*, n+1] \\
&\cong \int^{m, n} \text{Set}(Jn, X) \times \text{Set}(Jm, Y) \times \mathfrak{S}(m^*, n) \\
&\cong \int^m \text{Set}(Jm^*, X) \times \text{Set}(Jm, Y) \\
&\cong \text{colim}_m \text{Set}(Jm, X) \times \text{Set}(Jm, Y) \\
&\cong \int^m \text{Set}(Jm, X \times Y) \\
&\cong \exp(X \times Y) .
\end{aligned}$$

- As you might expect

$$\frac{d^k}{dX^k} \exp(X \times Y) \cong \exp(X \times Y) \times Y^k .$$

- Applying our antiderivative formula to $\exp(X \times Y)$ we obtain

$$R(X, Y) \cong \exp(X \times Y) \times \sum_{k \geq 0} (-1)^k (X^{k+1} / \mathfrak{S}_{k+1}) \times Y^k .$$

17 Review of some q concepts

1. The n -th q -integer

$$[n]_q = \frac{q^n - 1}{q - 1} = 1 + q + q^2 + q^3 + \cdots + q^{n-1}$$

is the cardinality of projective space of dimension n over the field $\mathbb{F}_q$ of cardinality q . That is, it is the number of one-dimensional subspaces of $\mathbb{F}_q^n$. Note that $[n]_1 = n$.

2. The q -factorial

$$[n]_q! = [n]_q [n-1]_q \cdots [2]_q [1]_q$$

is the number of complete flags $0 < V_1 < V_2 < \cdots < V_n = V$ in an n -dimensional $\mathbb{F}_q$ -vector space V .

3. Put

$$\phi_n(q) = (q^n - 1)(q^{n-1} - 1) \cdots (q - 1) = (q - 1)^n [n]_q! .$$

If V is an n -dimensional $\mathbb{F}_q$ -vector space then the number of linear bijections from V to itself is

$$\# \text{GL}(n, q) = (q^n - 1)(q^n - q) \cdots (q^n - q^{n-1}) = q^{\frac{1}{2}n(n-1)} \phi_n(q) .$$

4. The q -binomial coefficient

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{[n]_q!}{[k]_q![n-k]_q!} = \frac{\phi_n(q)}{\phi_k(q)\phi_{n-k}(q)}$$

is the number of k -dimensional subspaces of an n -dimensional $\mathbb{F}_q$ -vector space.

5. The q -derivative of a function $f(x)$ is defined by

$$(D_q f)(x) = \frac{f(qx) - f(x)}{qx - x}.$$

6. The q -integral [17] of a function $f(x)$ is defined by

$$(I_q f)(x) = \sum_{k \geq 0} (1 - q)xq^k f(q^k x).$$

These definitions make sense and can be applied even when q is not a prime power.

Example 17.1.

$$(D_q f(a \sim))(x) = \frac{f(qax) - f(ax)}{qx - x} = a \frac{f(qax) - f(ax)}{qax - ax} = a(D_q f)(ax).$$

Example 17.2. Suppose $f(x) = x^n$. Then

$$(D_q f)(x) = \frac{(qx)^n - x^n}{qx - x} = \frac{(q^n - 1)x^n}{(q - 1)x} = [n]_q x^{n-1}.$$

Example 17.3. $D_q(f + g) = D_q f + D_q g$.

Let us write $f \cdot g$ for the pointwise product $(f \cdot g)(x) = f(x)g(x)$ of functions. We have the q -Leibniz formula:

Example 17.4. $D_q(f \cdot g) = D_q f \cdot g(q \sim) + f \cdot D_q g = D_q f \cdot g + f(q \sim) \cdot D_q g$.

Proposition 17.5. $D_q I_q f = f$.

Proof. We can directly calculate

$$\begin{aligned} D_q(I_q(f))(x) &= \frac{I_q(f)(qx) - I_q(f)(x)}{qx - x} \\ &= \sum_{k \geq 0} (q^k f(q^k x) - q^{k+1} f(q^{k+1} x)) \\ &= f(x) - qf(qx) + qf(qx) - q^2 f(q^2 x) + q^2 f(q^2 x) - q^3 f(q^3 x) + \dots \\ &= f(x) \end{aligned}$$

since all but the first term cancel. □

The q -exponential function is defined by

$$e_q(x) = \sum_{n \geq 0} \frac{x^n}{[n]_q!} = \sum_{n \geq 0} (q-1)^n \frac{x^n}{\phi_n(q)} .$$

Using Examples 17.1, 17.2 and (the infinite version of) 17.3, we have that the q -exponential is fixed under q -differentiation:

$$D_q e_q = e_q . \quad (17.47)$$

Using Example 17.4, we have an antiderivative formula:

Proposition 17.6. *Suppose $D_q f_n = f_{n+1}$ and $D_q g_{n+1} = g_n(q \sim)$ for all $n \geq 0$. Then*

$$f_0 \cdot g_0(q \sim) = D_q \sum_{k \geq 0} (-1)^k f_k \cdot g_{k+1} .$$

Apply this to $f_n(x) = (D_q^n f)(x)$ and $g_0(x) = 1$, $g_n(x) = q^{\frac{1}{2}n(n-1)} \frac{x^n}{[n]_q!}$ to obtain the following formula for a q -antiderivative for any function f .

$$\int_q f(x) dx = \sum_{n \geq 0} (-1)^n q^{\frac{1}{2}n(n+1)} (D_q^n f)(x) \frac{x^{n+1}}{[n+1]_q!} \quad (17.48)$$

From Example 17.2, we deduce:

Proposition 17.7. *The q -derivative of*

$$f(x) = \sum_{n \geq 0} a_n \frac{x^n}{[n]_q!}$$

is

$$(D_q f)(x) = \sum_{n \geq 0} a_{n+1} \frac{x^n}{[n]_q!} .$$

18 Representations of the general linear groupoid over $\mathbb{F}_q$

Let us now make a connection with our paper [23] in which we studied the q -general linear groupoid $\mathcal{GL}_q$ of finite $\mathbb{F}_q$ -vector spaces with linear bijections. The external tensor product $M \otimes N$ of $[\mathcal{GL}_q, \text{Vect}_{\mathbb{C}}]$ was defined in Section 2 of [23] by

$$(M \otimes N)(A) = \bigoplus_{C \leq A} M(C) \otimes N(A/C) . \quad (18.49)$$

In Section 3, we looked at the *dimension series*

$$(\dim M)(x) = \sum_{n \geq 0} \dim M(\mathbb{F}_q^n) \frac{x^n}{\phi_n(q)} \quad (18.50)$$

of functors $M : \mathcal{GL}_q \rightarrow \text{Vect}_{\mathbb{C}}$ landing in finite dimensional complex vector spaces. In Section 3 of [23], it was shown (involving the q -binomial coefficient) that

$$\dim(M \otimes N) = \dim M \cdot \dim N . \quad (18.51)$$

Now we would like to modify this a little and define a *q -Taylor dimension series* by

$$(\text{dit} M)(x) = \sum_{n \geq 0} \dim M(\mathbb{F}_q^n) \frac{x^n}{[n]_q!} = (\dim M)((q-1)x) \quad (18.52)$$

We also have

$$\text{dit}(M \otimes N) = \text{dit} M \cdot \text{dit} N . \quad (18.53)$$

Encouraged by Proposition 17.7, we define the derivative functor $M' : \mathcal{GL}_q \rightarrow \text{Vect}_{\mathbb{C}}$ of M by

$$M'(A) = M(A \oplus \mathbb{F}_q) \quad (18.54)$$

to obtain

$$\text{dit} M' = D_q \text{dit} M . \quad (18.55)$$

The formula (18.49), with $\text{Vect}_{\mathbb{C}}$ replaced by any finitely-cocomplete symmetric monoidal category $\mathcal{V}$ and direct sum replaced by coproduct, defines a monoidal structure on $[\mathcal{GL}_q, \mathcal{V}]$. We also define the derivative of a functor by equation (18.54) in the general case. When $\mathcal{V} = \text{Set}$, we obtain the *external product* $E \times F$ of q -species; see Section 2 of [23]. The free-complex-vector-space functor $\text{Set} \rightarrow \text{Vect}_{\mathbb{C}}$ induces a strong monoidal functor between the functor categories; so

$$\mathbb{C}(E \times F) \cong \mathbb{C}E \otimes \mathbb{C}F .$$

The *k -Grassmannian q -species* is the functor $\text{Gr}_k : \mathcal{GL}_q \rightarrow \text{Set}$ whose value at $A \in \mathcal{GL}_q$ is the set $\text{Gr}_k(A) = \left[\begin{smallmatrix} A \\ k \end{smallmatrix} \right]_q$ of k -dimensional subspaces of A ; see page 912 of [23]. Item 4 above can be rephrased as the equation:

$$\#\text{Gr}_k(\mathbb{F}_q^n) = \left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]_q .$$

An alternative description of $\text{Gr}_k(\mathbb{F}_q^n)$ is as set of $k \times n$ matrices of rank k modulo the elementary row operations (obtained by multiplying by an invertible $k \times k$

matrix). Write $\text{rk}_k(n)$ for the set of $k \times n$ matrices over $\mathbb{F}_q$ of rank n . It is acted on by $\text{rk}_k(k) = \text{GL}(k, q)$ via matrix multiplication. So $\text{Gr}_k(\mathbb{F}_q^n) \cong \text{rk}_k(n)/\text{GL}(k, q)$.

A way of obtaining representatives for the orbits of $k \times n$ matrices of rank k is to take the corresponding reduced row echelon form. For this, we need the order-preserving, first-element-preserving, injective functions $\xi : \mathbf{k} \rightarrow \mathbf{n}$ to tell us where the pivotal 1s should be placed, namely, at the $(i, \xi(i))$ entry. Let

$$\gamma(\xi) = \{(i, j) \in \mathbf{k} \times \mathbf{n} : \xi(i) < j \notin \text{im} \xi\} ,$$

the set of positions where arbitrary elements of $\mathbb{F}_q$ can be put. For each ξ and function $u \in \mathbb{F}_q^{\gamma(\xi)}$, we obtain a $k \times n$ matrix (ξ, u) of rank k defined by

$$(\xi, u)_{ij} = \begin{cases} 1 & \text{if } j = \xi(i) \\ u(i, j) & \text{if } (i, j) \in \gamma(\xi) \\ 0 & \text{otherwise .} \end{cases}$$

Let $\Xi_{k,n}$ be the set of pairs (ξ, u) where ξ and u are as above. We obtain a bijection

$$\Xi_{k,n} \cong \text{rk}_k(n)/\text{GL}(k, q) \quad (18.56)$$

taking the pair (ξ, u) to the orbit $[\xi, u]$ of the matrix (ξ, u) .

We can define a q -species $\Xi_k : \mathcal{GL}_q \rightarrow \text{Set}$ where $\Xi_k(\mathbb{F}_q^n) = \Xi_{k,n}$ and, for $h : \mathbb{F}_q^n \xrightarrow{\cong} \mathbb{F}_q^n$ regarded as an invertible $n \times n$ matrix, the matrix $\Xi_k(h)(\xi, u)$ is in the row reduced echelon form obtained by Gaussian elimination from the matrix $(\xi, u)h$. For an arbitrary $A \in \mathcal{GL}_q$, we have

$$\Xi_k(A) = \sum_{n \geq 0} \mathcal{GL}_q(\mathbb{F}_q^n, A) \times_{\text{GL}(n, q)} \Xi_k(\mathbb{F}_q^n) .$$

This is designed so that we have a q -species isomorphism

$$\text{Gr}_k \cong \Xi_k . \quad (18.57)$$

We have an injective function $\Lambda_1 : \Xi_{k,n} \times \mathbb{F}_q^k \rightarrow \Xi_{k,n+1}$ defined by

$$\Lambda_1(\xi, u, \alpha) = (\xi', u')$$

where $\xi'(i) = \xi(i)$ for $i \in \mathbf{k}$, and $u'(i, j) = u(i, j)$ for $j \in \mathbf{n}$ and $u'(i, n+1) = \alpha_i$.

We also have an injective function $\Lambda_2 : \Xi_{k-1,n} \rightarrow \Xi_{k,n+1}$ defined by

$$\Lambda_2(\zeta, v) = (\zeta', v)$$

where $\xi'(i) = \xi(i)$ for $i \in \mathbf{k} - \mathbf{1}$ while $\xi'(k) = n$. That is, $\Lambda_1(a, \alpha)$ and $\Lambda_2(b)$ are respectively

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & \alpha_1 \\ a_{21} & a_{22} & \dots & a_{2n} & \alpha_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{k-11} & a_{k-12} & \dots & a_{k-1n} & \alpha_{k-1} \\ a_{k1} & a_{k2} & \dots & a_{kn} & \alpha_k \end{bmatrix} \quad \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1n} & 0 \\ b_{21} & b_{22} & \dots & b_{2n} & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ b_{k-11} & b_{k-12} & \dots & b_{k-1n} & 0 \\ 0 & 0 & \dots & 0 & 1 \end{bmatrix}$$

for $(a, \alpha) \in \Xi_{k,n} \times \mathbb{F}_q^k$ and $b \in \Xi_{k-1,n}$.

Proposition 18.1. *The functions Λ_1 and Λ_2 provide a bijection*

$$\Xi_{k,n} \times \mathbb{F}_q^k + \Xi_{k-1,n} \cong \Xi_{k,n+1}$$

natural in n .

Proof. Λ_1 and Λ_2 are injective, have disjoint images, and are jointly surjective. $\square$

It follows from (18.57) that we have a natural isomorphism

$$\mathrm{Gr}'_k \cong \mathrm{Gr}_k \times \mathbb{F}_q^k + \mathrm{Gr}_{k-1} \quad (18.58)$$

for the derivative of the k -Grassmannian which is an objective form of the equation

$$\begin{bmatrix} n+1 \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ k \end{bmatrix}_q q^k + \begin{bmatrix} n \\ k-1 \end{bmatrix}_q.$$

Each functor $M : \mathcal{GL}_q \rightarrow \mathrm{Vect}_{\mathbb{C}}$ determines a functor $M^{\$} : \mathcal{GL}_q \rightarrow \mathrm{Vect}_{\mathbb{C}}$ defined by $M^{\$}(A) = A \cdot M(A)$ (the direct sum of $\#A$ copies of $M(A)$).

We have the following version of the Leibniz rule for differentiation of representations.

Proposition 18.2. $(M \otimes N)' \cong M^{\$} \otimes N' \oplus M' \otimes N$

Proof. We have the following calculation

$$\begin{aligned} (M \otimes N)'(\mathbb{F}_q^n) &= (M \otimes N)(\mathbb{F}_q^n \oplus \mathbb{F}_q) \\ &= \bigoplus_{C \leq \mathbb{F}_q^n \oplus \mathbb{F}_q} M(C) \otimes N(\mathbb{F}_q^n \oplus \mathbb{F}_q / C) \\ &\cong \bigoplus_{k \geq 0, C \in \mathrm{Gr}_k(\mathbb{F}_q^n)} C \cdot M(C) \otimes N(\mathbb{F}_q^n / C \oplus \mathbb{F}_q) \\ &\quad \oplus \bigoplus_{k > 0, D \in \mathrm{Gr}_{k-1}(\mathbb{F}_q^n)} M(D \oplus \mathbb{F}_q) \otimes N(\mathbb{F}_q^n / D) \\ &\cong (M^{\$} \otimes N')(\mathbb{F}_q^n) \oplus (M' \otimes N)(\mathbb{F}_q^n) \end{aligned}$$

where the third step uses the isomorphism 18.58. $\square$

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